MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3360 CONCENTRATE STORAGE & REPULP

DESIGN CALCULATION OF GRADE SLAB

AS BUILT

MD-216-3360-EG-CV-CAL-1232

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

TABLE OF CONTENTS

		PAGE NO
DEFINITIONS and ABBREVIATIONS		3
1	INTENT	4
2	SCOPE	4
3	Design of Foundation for Concentrate Reclaim Weighfeeder	5
	3.1 Design of Foundation mkd. F1	5
	3.2 Design of Foundation mkd. F2	8
	3.3 Design of Foundation mkd. F3	11
4	Design of Retaining Wall	15
	4.1 Design of Retaining Wall around Truck loading Ramp inside Concentrate Storage Building	15
	4.2 Design of Retaining Wall around Ramp inside Concentrate Filtration Building	19
REFERENCES		23
ATTACHMENT 1	CONCENTRATE FILTRATION LOAD PLAN MECHANICAL INPUT	24
ATTACHMENT 2	LAYOUT & LOAD DATA OF CONCENTRATE RECLAIM WEIGHFEEDER	26
ATTACHMENT 3	STAAD Input & Output File of Foundation Mkd. F1	29
ATTACHMENT 4	STAAD Input & Output File of Foundation Mkd. F2	44
ATTACHMENT 5	STAAD Input & Output File of Foundation Mkd. F3	53
ATTACHMENT 6	STAAD Input & Output File of WEIGHFEEDER SUPPORT STRUCTURE	61

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Foundation for Retaining Walls & Structure supporting Concentrate Reclaim Weighfeeder in Concentrate Storage area .

2. SCOPE

The document is prepared for Design Calculation of Foundations for Conecntrate Reclaim Weighfeeder in the area of Concentrate Storage. Three types of Open Foundation namely F1 to F3 are designed for all the supports.
Load data for foundation is attached with this document in attachement 1, 2 & 6.
Foundations are modelled & analysed in Staad-Pro. Loads from Load Data have been applied as loads to Foundations. As Seismic load is less then wind load
so foundations are designed for wind load. Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008 (as supported by STAAD-Pro).
Net Safe Bearing Capacity is obtained from Geotechnical report BH07 . Approximate value of 50T/sq.m is considered
Additionally, Retaining walls for Ramps at different locations of Storage area are also designed in this document.

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3 Design of Foundation for Concentrate Reclaim Weighfeeder

3.1 Design of Foundation mkd. F1

(Refer Load Data for Concentrate Reclaim Weighfeeder)

	222 & 224			223 & 225			226 & 228			227 & 229		
LOAD ITEM	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton	Fx Ton	Fy Ton	Fz Ton
1 DL	-0.8	-2.7	0.0	0.8	-2.7	0.0	-0.2	-0.9	0.0	0.2	-0.9	0.0
2 LL	-3.0	-28.0	-0.4	3.0	-28.0	-0.4	0.0	-5.0	0.0	0.0	-5.0	0.0

FOUNDATION SIZE (F1) 4.00m X 3.50m
DEPTH OF THE FOUNDATION 2.00m from FFL
THE WT OF SOIL +SELF WT 4x3.5x2x2 = 56 T

WORKING STRESS DESIGN	222 & 224			223 & 225			226 & 228			227 & 229		
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)
DL+LL	-3.80	-30.70	-0.40	3.80	-30.70	-0.40	-0.20	-5.93	0.00	0.20	-5.93	0.02

SLIDING AND OVERTURNING CHECK

	Load at bottom of foundation					OVERTURNING CHECK		SLIDING CHECK, m = 0.4	
LOAD CASES	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	129.25	0.0	2.0	-0.78	61.94	4.17	115.52	21542.13	66.11

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 20 T/m^2 IN LC 10. HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

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Foundation has been designed with the Design strength Load Combinations of Staad File (LC 20)

REINFORCEMENT CALCULATION OF FOUNDATION:

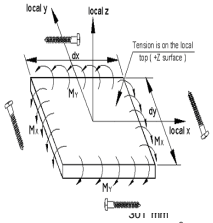
Foundation Thickness =	400	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	8.61	T-M	Corresponding Mxy	-1.98	T-M	*(Plate No. - 237 , L/C No. - 20 1)*
Min. Mx	-3.173	T-M	Corresponding Mxy	-1.30	T-M	*(Plate No. - 232 , L/C No. - 20 1)*
Max. My	5.272	T-M	Corresponding Mxy	-5.60	T-M	*(Plate No. - 322 , L/C No. - 20 1)*
Min. My	-5.208	T-M	Corresponding Mxy	-0.82	T-M	*(Plate No. - 3 , L/C No. - 20 1)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	8.61 + 1.979 =	10.589	T-M
My1 =	My + abs(Mxy) =	5.272 + 5.602 =	10.874	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	8.61 + 0.743 =	9.353	T-M
My2 =	My+abs(Mxy^2/Mx) =	5.272 + 3.645 =	8.917	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 10.589 T-m
Myd = 10.874 T-m

ALONG X - DIRECTION :

Mu = 10.589 T-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 10.589X10^7/(1000X301^2) = 1.169 N/mm^2 < 6.9 N/mm^2
Pt = 0.0032 Reqd. Ast = 0.0032X1000X301/100 = 9.63 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 10.874 T-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 10.874X10^7/(1000X317^2) = 1.082 N/mm^2 < 6.9 N/mm^2
Pt = 0.0029 Reqd. Ast = 0.0029X1000X317/100 = 9.19 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05 cm2) at Horizontal

Total Provide Ast = 10.05 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-3.173 - 1.3 =	-4.473	T-M
My1 =	My - abs(Mxy) =	-5.208 - 0.821 =	-6.029	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-3.173 - 0.325 =	-3.498	T-M
My2 =	My-abs(Mxy^2/Mx) =	-5.208 - 0.212 =	-4.996	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -4.473 T-M
Myd = -6.029 T-M

ALONG X - DIRECTION :

Mu = 4.473 T-m
Φ*Kn = Mu/bd^2 deff. = 307 mm
Mu/bd^2 = 4.473X10^7/(1000X307^2) = 0.475 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X307/100 = 3.99 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 6.029 T-m
Φ*Kn = Mu/bd^2 deff. = 319 mm
Mu/bd^2 = 6.029X10^7/(1000X319^2) = 0.592 N/mm^2 < 6.9 N/mm^2
Pt = 0.0016 Reqd. Ast = 0.0016X1000X319/100 = 5.1 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

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SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.674 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 240 , L/C No. - 20 1)*
Vu = 20.692 T

Provided Pt = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X20.692X10^4X307/4.473/10^7)X1000X307/10^4
= 31.506 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X307/10^4
= 30.782 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X307/10^4
= 52.671 T
HENCE, Vc = 30.782 T
Φt*Vc = 0.75X30.782 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 23.087 T > **20.692T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.073 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 35 , L/C No. - 20 1)*
Vu = 2.329 T

Pt provided = 0.0018
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0018X2.329X10^4X319/6.029/10^7)X1000X319/10^4
= 30.316 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0018)X1000X319/10^4
= 31.172 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X319/10^4
= 54.73 T
HENCE, Vc = 30.316 T
Φt*Vc = 0.75X30.316 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 22.737 T > **2.329T , Hence O.K.**

DESIGN OF PEDESTAL: REFER TO STAAD FILE OF FOUNDATION MKD. F1

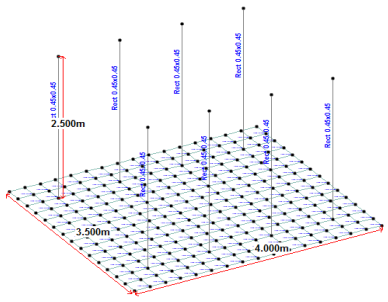
Size of Column = 450 X 450 mm^2

Reqired area of Reinforcement = 2025 mm^2

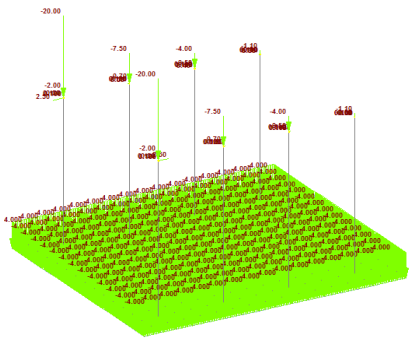
Reinforcement Provided = 4 no. 16 dia bars + 4 no. 20 dia bars

Area of Reinforcement Provided = 2060.9 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + LIVE+ SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 1.6x 1.8 T/m2 = 2.88 T/m^2
SOIL+SURCHARGE LOAD = 3.88 T/m^2

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3.2 Design of Foundation mkd. F2 (Refer Attachment 6 for Foundation Load Data)

	C-1			C-2		
LOAD ITEM	Fx (T)	Fy (T)	Fz (T)	Fx (T)	Fy (T)	Fz (T)
1 DL	0.00	-2.00	0.00	0.00	-1.50	0.00
2 LL	0.00	-3.50	0.00	0.00	-2.50	0.00
3 WL1(+X)	0.10	0.21	0.00	0.10	-0.21	0.00
4 WL2(-X)	-0.10	-0.21	0.00	-0.10	0.21	0.00
5 WL3(+Z)	0.50	-0.70	0.50	0.00	0.35	0.30
6 WL4(-Z)	-0.50	0.70	-0.50	0.00	-0.35	-0.30

FOUNDATION SIZE 1.20m X 2.50m
DEPTH OF THE FOUNDATION 2.00m (below FFL)
THE WT OF SOIL +SELF WT 1.2X2.5X2XUNIT WT = 12 T

WORKING STRESS DESIGN

	C-1			C-2			Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	FX (T)	FY(T)	FZ(T)	V (T)	Hx (T)	MZ (T-m)	Hz (T)	MX (T-m)
DL+LL	0.00	-5.50	0.00	0.00	-4.00	0.00	21.50	0.00	0.00	0.00	1.13
DL+0.75*LL+0.75*WL1	0.08	-4.47	0.00	0.08	-3.53	0.00	20.00	0.15	0.00	0.00	1.00
DL+0.75*LL+0.75*WL2	-0.08	-4.78	0.00	-0.08	-3.22	0.00	20.00	-0.15	0.00	0.00	1.47
DL+0.75*LL+0.75*WL3	0.38	-5.15	0.38	0.00	-3.11	0.23	20.26	0.38	1.20	0.60	2.28
DL+0.75*LL+0.75*WL4	-0.38	-4.10	-0.38	0.00	-3.64	-0.23	19.74	-0.38	1.20	-0.60	1.10
0.6*DL+WL1	0.10	-0.99	0.00	0.10	-1.11	0.00	14.10	0.20	0.00	0.00	0.49
0.6*DL+WL2	-0.10	-1.41	0.00	-0.10	-0.69	0.00	14.10	-0.20	0.00	0.00	0.94
0.6*DL+WL3	0.50	-1.90	0.50	0.00	-0.55	0.30	14.45	0.50	1.60	0.80	2.01
0.6*DL+WL4	-0.50	-0.50	-0.50	0.00	-1.25	-0.30	13.75	-0.50	1.60	-0.80	1.56

Check for overturning and sliding:

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	MX (Tm)	Hz (T)	MZ (Tm)	F.O.S. MX	F.O.S. MZ	F.O.S. Hx	F.O.S. Hz
DL+LL	21.50	0.00	1.13	0.00	0.00	23.89	-	-	-
DL+0.75*LL+0.75*WL1	20.00	0.15	1.00	0.00	0.00	24.97	-	53.33	-
DL+0.75*LL+0.75*WL2	20.00	-0.15	1.47	0.00	0.00	16.96	-	53.33	-
DL+0.75*LL+0.75*WL3	20.26	0.38	2.28	0.60	1.20	11.12	10.13	21.61	13.51
DL+0.75*LL+0.75*WL4	19.74	-0.38	1.10	-0.60	1.20	22.49	9.87	21.05	13.16
0.6*DL+WL1	14.10	0.20	0.49	0.00	0.00	35.97	-	28.20	-
0.6*DL+WL2	14.10	-0.20	0.94	0.00	0.00	18.75	-	28.20	-
0.6*DL+WL3	14.45	0.50	2.01	0.80	1.60	8.98	5.42	11.56	7.23
0.6*DL+WL4	13.75	-0.50	1.56	-0.80	1.60	11.00	5.16	-	6.88

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m^2
MAXIMUM GROSS BEARING PRESSURE OBTAINED 10T/m^2 IN LC 101 . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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Foundation has been designed with the Design strength Load Combinations of Staad File (LC 200 - LC 205) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

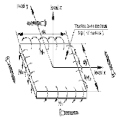
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	0.675	T-M	Corresponding Mxy	0.11	T-M	*(Plate No. - 121 , L/C No. - 203)*
Min. Mx	-0.011	T-M	Corresponding Mxy	-0.07	T-M	*(Plate No. - 86 , L/C No. - 204)*
Max. My	0.804	T-M	Corresponding Mxy	0.05	T-M	*(Plate No. - 126 , L/C No. - 203)*
Min. My	-0.853	T-M	Corresponding Mxy	-0.01	T-M	*(Plate No. - 93 , L/C No. - 201)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	0.675 + 0.105 =	0.78 T-M
My1 =	My + abs(Mxy) =	0.804 + 0.053 =	0.857 T-M
Mx2 =	Mx+abs(Mxy^2/My) =	0.675 + 0.014 =	0.689 T-M
My2 =	My+abs(Mxy^2/Mx) =	0.804 + 0.004 =	0.808 T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 0.78 T-m
Myd = 0.857 T-m

ALONG X - DIRECTION :

Mu = 0.78 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.78X10^7/(1000X207^2) = 0.182 N/mm^2 < deff. = 207 mm = 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X207/100 = 1.04 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 0.857 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.857X10^7/(1000X219^2) = 0.179 N/mm^2 < deff. = 219 mm = 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X219/100 = 1.1 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-0.011 - 0.071 =	-0.082 T-M
My1 =	My - abs(Mxy) =	-0.853 - 0.013 =	-0.866 T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-0.011 - 0.006 =	-0.017 T-M
My2 =	My-abs(Mxy^2/Mx) =	-0.853 - 0.015 =	-0.838 T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -0.082 T-M
Myd = -0.866 T-M

ALONG X - DIRECTION :

Mu = 0.082 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.082X10^7/(1000X207^2) = 0.019 N/mm^2 < deff. = 207 mm = 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X207/100 = 0.21 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 0.866 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 0.866X10^7/(1000X219^2) = 0.181 N/mm^2 < deff. = 219 mm = 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X219/100 = 1.1 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
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Outotec / L&T Revision No.:	0	1	2	3
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SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = -0.015 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 118 , L/C No. - 203)*
Vu = 0.311 T

Provided Pt = 0.0027
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0027X0.311X10^4X207/0.082/10^7)X1000X207/10^4
= 20.34 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4
= 20.544 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X207/10^4
= 35.514 T
HENCE, Vc = 20.34 T
Φt*Vc = 0.75X20.34 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.255 T **> 0.311T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = 0.082 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 117 , L/C No. - 203)*
Vu = 1.796 T

Pt provided = 0.0026
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0026X1.796X10^4X219/0.866/10^7)X1000X219/10^4
= 21.17 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4
= 21.698 KN
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X219/10^4
= 37.573 T
HENCE, Vc = 21.17 T
Φt*Vc = 0.75X21.17 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 15.878 T **> 1.796T , Hence O.K.**

PEDESTAL REINFORCEMENT

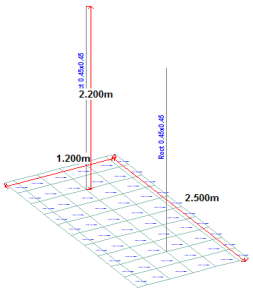
Size of Column = 450 X 450 mm^2

Reqired area of Reinforcement = 2025 mm^2

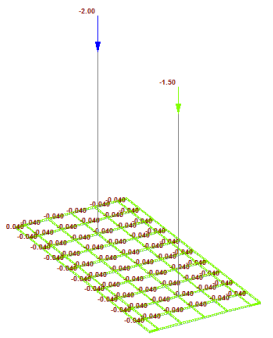
Reinforcement Provided = 4 no. 16 dia bars + 4 no. 20 dia bars

Area of Reinforcement Provided = 2060.9 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2
Soil Weight over the foundation = 1.7x 1.8 T/m2 = 3.06 T/m^2

SOIL+SURCHARGE LOAD = 4.06 T/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
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3.3 Design of Foundation mkd. F3

(Refer Attachment 6 for Foundation Load Data)

Load Detail	Node 3, 4		
	FX (T)	FY (T)	FZ (T)
1 DL	1.0	2.0	0.1
2 LL	2.2	3.7	-0.2
3 WL(+X)	-0.2	-0.2	0.0
4 WL(+Z)	-0.1	0.1	-0.2

Strength Load Combinations as per Cl. No. 2.3.2, SBC 301:2007 for the most critical support reactions as follows:

STRENGTH DESIGN	Node 3, 4		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4DL	1.4	2.8	0.12
1.4DL+1.7LL	5.1	9.1	-0.2
1.2D+1.0L+1.6W(+X)	3.0	5.8	-0.1
1.2D+1.0L+1.6W(+Z)	3.3	6.2	-0.4
0.9D+1.6W(+X)	0.5	1.48	0.1
0.9D+1.6W(+Z)	0.8	1.94	-0.2

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Node 3, 4		
Load Combinations	FX (T)	FY (T)	FZ (T)
DL+LL	3.2	5.7	-0.1
DL+0.75*LL+0.75*WLX	2.5	4.6	-0.1
DL+0.75*LL+0.75*WLZ	2.6	4.8	-0.2
0.6*DL+WLX	0.4	1.0	0.1
0.6*DL+WLZ	0.5	1.3	-0.1

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01
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DESIGN OF FOUNDATION MKD. F3

	ABOUT-X	ABOUT-Z
Size of column :	0.45m	0.45m
Size of fdn. raft :	1.65m	1.65m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.80 m	
Unit weight of conc :	2.5 T/m³	
Unit weight of soil :	1.8 T/m³	

SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

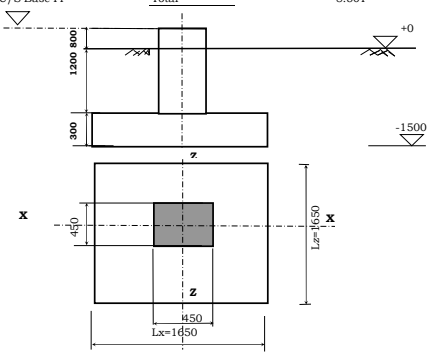
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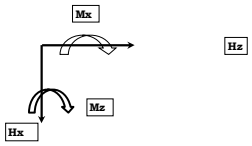
Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

Effective depth of slab along X direction = 217mm = 0.217m
Effective depth of slab along Z direction = 201mm = 0.201m
Clear projection along Z direction = 0.6m
Clear projection along X direction = 0.6m

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.65*1.65*0.3*2.5)+(0.45*0.45*2*2.5) = 3.05T
Weight of soil = ((1.65*1.65)-(0.45*0.45))*1.2*1.8 = 5.44T
U/S Base Pl Total = 8.50T





WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+LL	3.20	5.70	-0.11	0.0	0.00	14.20	3.20	0.26	-0.11	7.36
DL+0.75*LL+0.75*WLX	2.47	4.63	-0.06	0.0	0.00	13.13	2.47	0.14	-0.06	5.68
DL+0.75*LL+0.75*WLZ	2.60	4.84	-0.21	0.0	0.00	13.34	2.60	0.49	-0.21	5.97
0.6*DL+WLX	0.36	1.00	0.05	0.0	0.00	9.50	0.36	0.12	0.05	0.83
0.6*DL+WLZ	0.53	1.29	-0.15	0.0	0.00	9.79	0.53	0.34	-0.15	1.22

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
Load Case	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
DL+LL	14.20	3.20	7.36	-0.11	0.26	1.59	45.48	1.78	50.71
DL+0.75*LL+0.75*WLX	13.13	2.47	5.68	-0.06	0.14	1.91	75.93	2.13	84.68
DL+0.75*LL+0.75*WLZ	13.34	2.60	5.97	-0.21	0.49	1.84	22.57	2.05	25.17
0.6*DL+WLX	9.50	0.36	0.83	0.05	0.12	9.47	64.54	10.56	71.97
0.6*DL+WLZ	9.79	0.53	1.22	-0.15	0.34	6.63	23.86	7.39	26.60

CHECK FOR BEARING PRESSURE

THE NET SAFE BEARING CAPACITY OF SOIL IS 50T/m²
MAXIMUM GROSS BEARING PRESSURE OBTAINED 17T/m² IN LC 101 . HENCE IT IS SAFE.
REFER TO THE STAAD FILE ATTACHED.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3
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Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI

Foundation has been designed with the Design strength Load Combinations of Staad File (LC 200 - LC 205) (REFER STAAD MODEL)

REINFORCEMENT CALCULATION OF FOUNDATION:

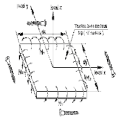
Foundation Thickness =	300	mm
Width =	1000	mm
Clear Cover =	75	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.706	T-M	Corresponding Mxy	0.06	T-M	*(Plate No. - 31 , L/C No. - 201)*
Min. Mx	-1.901	T-M	Corresponding Mxy	0.22	T-M	*(Plate No. - 82 , L/C No. - 201)*
Max. My	2.353	T-M	Corresponding Mxy	0.06	T-M	*(Plate No. - 31 , L/C No. - 201)*
Min. My	-0.448	T-M	Corresponding Mxy	0.22	T-M	*(Plate No. - 82 , L/C No. - 201)*

FOR OPPOSITE FACE REINFORCEMENT :

Mx1 =	Mx + abs(Mxy) =	2.706 + 0.057 =	2.763	T-M
My1 =	My + abs(Mxy) =	2.353 + 0.057 =	2.41	T-M
Mx2 =	Mx+abs(Mxy^2/My) =	2.706 + 0.001 =	2.707	T-M
My2 =	My+abs(Mxy^2/Mx) =	2.353 + 0.001 =	2.354	T-M



DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT (Bottom) :

Mxd = 2.763 T-m
Myd = 2.41 T-m

ALONG X - DIRECTION :

Mu = 2.763 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.763X10^7/(1000X207^2) = 0.645 N/mm^2 < 6.9 N/mm^2
Pt = 0.0017 Reqd. Ast = 0.0017X1000X207/100 = 3.52 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 2.41 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 2.41X10^7/(1000X219^2) = 0.502 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X219/100 = 2.85 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

FOR RETAINED FACE REINFORCEMENT :

Mx1 =	Mx - abs(Mxy) =	-1.901 - 0.222 =	-2.123	T-M
My1 =	My - abs(Mxy) =	-0.448 - 0.222 =	-0.67	T-M
Mx2 =	Mx-abs(Mxy^2/My) =	-1.901 - 0.11 =	-2.011	T-M
My2 =	My-abs(Mxy^2/Mx) =	-0.448 - 0.026 =	-0.422	T-M

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT (Top) :

Mxd = -2.123 T-M
Myd = -0.67 T-M

ALONG X - DIRECTION :

Mu = 2.123 T-m
Φ*Kn = Mu/bd^2 deff. = 207 mm
Mu/bd^2 = 2.123X10^7/(1000X207^2) = 0.495 N/mm^2 < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X207/100 = 2.69 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

ALONG Y - DIRECTION :

Mu = 0.67 T-m
Φ*Kn = Mu/bd^2 deff. = 219 mm
Mu/bd^2 = 0.67X10^7/(1000X219^2) = 0.14 N/mm^2 < 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X219/100 = 0.88 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal

Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.00

Global Project ID	900707
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Outotec / L&T Revision No.:	0	1	2	3
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SHEAR CHECK

(FROM FACTORED LOAD COMBINATION)

ALONG X - DIRECTION :

SQx = 0.148 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 144 , L/C No. - 201)*

Vu = 3.064 T

Provided Pt = 0.0027

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0027X3.064X10^4X207/2.123/10^7)X1000X207/10^4

= 19.878 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0027)X1000X207/10^4

= 20.544 T

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X207/10^4

= 35.514 T

HENCE, Vc = 19.878 T

Φt*Vc = 0.75X19.878 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 14.909 T **> 3.064T , Hence O.K.**

ALONG Y - DIRECTION :

SQy = -0.092 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 145 , L/C No. - 201)*

Vu = 2.015 T

Pt provided = 0.0026

Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*

[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1

= (0.16X1X35^0.5+17X0.0026X2.015X10^4X219/0.67/10^7)X1000X219/10^4

= 21.367 T

[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff

= (0.16X1X35^0.5+17X0.0026)X1000X219/10^4

= 21.698 KN

[c] = 0.29*λ*fc^0.5*b*d.eff

= 0.29X1X5.92X1000X219/10^4

= 37.573 T

HENCE, Vc = 21.367 T

Φt*Vc = 0.75X21.367 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*

= 16.025 T **> 2.015T , Hence O.K.**

PEDESTAL REINFORCEMENT

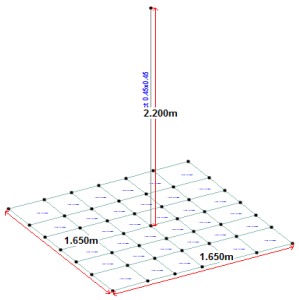
Size of Column = 450 X 450 mm^2

Reqired area of Reinforcement = 2025 mm^2

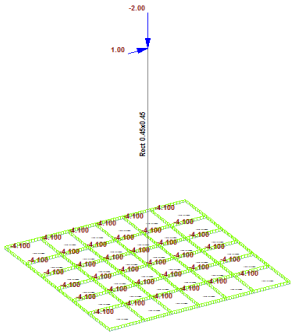
Reinforcement Provided = 4 no. 16 dia bars + 4 no. 20 dia bars

Area of Reinforcement Provided = 2060.9 mm^2

STAAD MODEL OF RAFT WITH PLATE ELEMENT



STAAD MODEL OF RAFT WITH DEAD + SURCHARGE LOAD



Surcharge Load on Foundation = 1 T/m^2

Soil Weight over the foundation = 1.7x 1.8 T/m2 = 3.06 T/m^2

SOIL+SURCHARGE LOAD = 4.06 T/m^2

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4 Design of Retaining Wall

4.1 Design of Retaining Wall around Truck loading Ramp inside Concentrate Storage Building

Retaining Wall of 1m Width & 4.5m high is modelled in Staad-Pro with relevant loads. It supports Ramp at a height of approximately 3m from base. Foundation Base of Retaining wall is at 1.5m below FGL.
There is compacted structural fill below 150 thk slab of Ramp. Supports are provided as plate mat with Modulus of subgrade reaction as 5400 T/m^2/m as per Geotechnical Report of BH-07

Load Calculation :

a) Dead Load (DL):

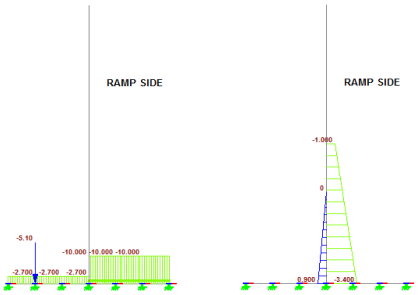
i) Self-Weight of Retaining Wall including Base Raft

ii) Wall Pressure due to compacted earthfill from base upto 2.9m height (applied as triangular distribution on wall in Staad)

Surcharge on Ramp =			3 T/m^2
Since the material stored is soil,	(as per dwg no. MD-216-3350-EG-ME-GAD-1053)		
Angle of internal friction, ϕ considered = 30° ;	k = (1-sinϕ)/(1+sinϕ) =		0.33
Self weight of Material Storage =	2.7* 2.5	=	6.75 T/m^2
Storage height considered from latest Tekla Model			
Pressure on Heel Slab at wall base = (excl. raft weight)			9.75 T/m^2
Pressure on Heel Slab at cantilever edge =			9.75 T/m^2
Pressure on Toe Slab:			
Weight of Soil over Toe Slab =	(2-0.4)*1.8 =		2.88 T/m^2

b) Horizontal Load due to material storage (H):

Lateral wall Pressure at top =	0.33*3	=	0.99 T/m^2
Lateral Wall Pressure at wall base due to compacted earthfill on Ramp Side=	0.33 * (6.75 + 3)=		3.22 T/m^2
Lateral earth Pressure on other side =	2.88*0.33 =		1.0 T/m^2



DEAD LOAD (loads are in T/m) HORIZONTAL STORAGE LOAD (loads are in T/m)

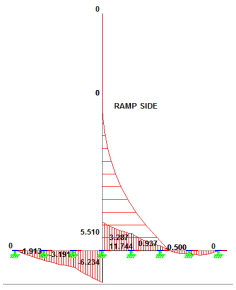
c) Seismic Load (EL):

Seismic Parameters : (Calculated according to Chapter 7, SBC 303)

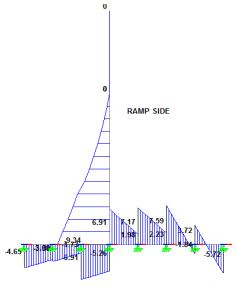
Seismic Design Category =	B	
Site Class =	B	
Seismic Coeffecient, Ss =	0.1 g	(From Civil-Structural Design Criteria)
S_1 =	0.02 g	
Occupancy importance factor, I =	1	(From Table 9.5, SBC 301:2007)
Response modification factor, R =	3	(From Table 10.2, SBC 301:2007)
Height of the Wall supporting material, h =	2.75 m	
Earthquake Load, Combined resultant Active Force, P =	1/2 * γ*H^2 *k_A * (1-k_v) *cos^{2}Θ* /cos^{2}Θ x cos ζ	(eqn. 7-1, Cl. No. 7.2.6, SBC 303:2008)
k_h = Horizontal Gound Acceleration =	0.1g =	
k_v = Vertical Gound Acceleration =	0.1g *2/3 = 0.066g	
ζ =	tan^{-1} (k_h/ 1-k_v) = 0.107	(eqn. 7-2, Cl. No. 7.2.6, SBC 303:2008)
Θ =	0	
Θ * = Θ + ζ = ζ :	cos ζ = 0.995	

Combined resultant Active Force, P
(0.5*(1.8*2.75+9.75)*2.75*0.33 *0.995*(1-0.067) = 5.08 T/m

Retaining wall has been designed for the following Load Combinations: (Table 5.3.1, SBC 304:2018) and ACI 318 : 2014
LOAD COMB 20 1.2*DL + 1.6*H
LOAD COMB 21 1.2DL+1.0EL
LOAD COMB 22 0.9DL+1.0EL + 1.6*H
Retaining wall has been designed for the following Bending Moment & Shear Force for LC- 0.9DL+1.0EL + 1.6*H (Most Critical):



BENDING MOMENT DIAGRAM



SHEAR FORCE DIAGRAM

(loads are in T-m)

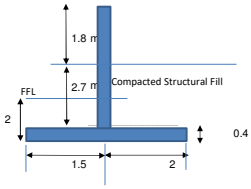
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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Stability Check for Retaining Wall

Lateral Wall Pressure at FFL = 3.3 T/m^2
Width of Heel Slab from Center of Wall = 2 m



Combined resultant Active Force, P = 5.08 T/m
Overturning Moment = 5.08 * (0.6* 2.75) = 8.4 T-m
(Considering, Combined force acts at 0.6 times height of wall, as per Cl. No. 7.2.6, SBC 303)

Average Thickness of Wall = 400 mm
Thickness of Raft = 400 mm
Self weight of Retaining Wall = 8 T/m

Restoring Forces : (Passive Earth Pressure not considered for simplicity)

i) Weight of Stored Material = 2.7*2.5 = 6.75 T/m
ii) Load due to self weight of hopper column = 0.5*0.5*2.0*2.5 = 1.25 T/m
iii) Load due to soil wt. on other side of ramp = 2*1.5* 1.8 = 5.40 T/m
Self Weight of Wall restoring moment = 8*1.5= 12.00 T-m (Restoring moment)

Total Weight = 21.4 T/m
Coeffecient of Friction between Concrete & Soil, μ = 0.4
Restoring Horizontal Force = 0.4*21.4 = 8.56 T/m

Moment due to Material = (2/2*+1.5)*(2.7*2)*2.5 = 33.75 T-m/m
Total Restoring Moment = 45.75 T-m/m
For the combined effect of static & earthquake load ,
FOS against Overturning = 45.75/8.382= 5.46 **> 1.2 - OK** (Refer Cl. No. 7.2.6, SBC 303)
FOS against Sliding = 8.56/5.08 = 1.69 **> 1.2 - OK** (Refer Cl. No. 7.2.6, SBC 303)

The Wall is Safe against Overturning & Sliding

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REINFORCEMENT CALCULATION OF WALL : (REFER STAAD OUTPUT FOR LC 20 TO 22)

Wall Thickness =	400	mm
Width =	1000	mm
Clear Cover =	50	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 5 T-m (ASSUMED)
Myd = 5 T-m

ALONG X - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X326^2) = 0.47 N/mm^2 < deff. = 326 mm 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X326/100 = 4.24 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0021

ALONG Y - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X342^2) = 0.427 N/mm^2 < deff. = 342 mm 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X342/100 = 3.76 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Vertical
Total Provide Ast = 6.7 cm2 at Vertical **O.K.**
Total Provide Pt = 0.002
Minimum Reinforcement ratio for both faces = 0.0025 (Cl. No. 11.6, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

Mxd = 5 T-M (ASSUMED)
Myd = 10 T-M (Refer Staad Output at face of raft) (LC -22 - 0.9DL+1.0EL + 1.6*H)

ALONG X - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X326^2) = 0.47 N/mm^2 < deff. = 326 mm 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X326/100 = 4.24 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0021

ALONG Y - DIRECTION :

Mu = 10 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 10X10^7/(X342^2) = 0.855 N/mm^2 < deff. = 342 mm 6.9 N/mm^2
Pt = 0.0023 Reqd. Ast = 0.0023XX342/100 = 7.87 cm^2
Provided **16** mm @ **150** C/C (Provide Ast = 13.4 cm2) at Vertical
Total Provide Ast = 13.4 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0039

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)
ALONG X - DIRECTION :

Vu = 8 T (Refer Staad Output at distance d from face of raft; LC - 22)

Provided Pt = 0.0039
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0039X8X10^4X342/10/10^7)X1000X342/10^4
= 32.993 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0039)X1000X342/10^4
= 34.64 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X342/10^4
= 58.676 T
HENCE, Vc = 32.993 T
Φt*Vc = 0.75X32.993 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 24.745 T **> 8T , Hence O.K.**

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REINFORCEMENT CALCULATION OF WALL RAFT (REFER STAAD OUTPUT FOR LC 20 TO 22)

Slab Thickness =	400	mm
Width =	1000	mm
Clear Cover =	50	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 6 T-m (Refer Staad Output at face of wall)(LC NO. 22)
Myd = 5 T-m (ASSUMED)

ALONG X - DIRECTION :

Mu = 6 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 6X10^7/(1000X342^2) = 0.513 N/mm^2 deff. = 342 mm < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast = 0.0014X1000X342/100 = 4.79 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.002

ALONG Y - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X326^2) = 0.47 N/mm^2 deff. = 326 mm < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X326/100 = 4.24 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0021
Minimum Reinforcement ratio for both faces = 0.0018 (Cl. No. Table 8.6.1.1, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

-0.036

Mxd = 5 T-M (Refer Staad Output at face of wall)(LC NO. 22)
Myd = 5 T-M (ASSUMED)

ALONG X - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X342^2) = 0.427 N/mm^2 deff. = 342 mm < 6.9 N/mm^2
Pt = 0.0011 Reqd. Ast = 0.0011X1000X342/100 = 3.76 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.002

ALONG Y - DIRECTION :

Mu = 5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 5X10^7/(1000X326^2) = 0.47 N/mm^2 deff. = 326 mm < 6.9 N/mm^2
Pt = 0.0013 Reqd. Ast = 0.0013X1000X326/100 = 4.24 cm^2
Provided **16** mm @ **300** C/C (Provide Ast = 6.7 cm2) at Horizontal
Total Provide Ast = 6.7 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0021

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)
ALONG X - DIRECTION :

Vu = 7 T (Refer Staad Output at a distance d from face of wall)(LC - 22 - 0.9DL+1.0EL + 1.6*H)

Provided Pt = 0.002
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.002X7X10^4X342/5/10^7)X1000X342/10^4
= 32.93 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.002)X1000X326/10^4
= 31.967 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X326/10^4
= 55.931 T
HENCE, Vc = 31.967 T
Φt*Vc = 0.75X31.967 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 23.975 T > **7T , Hence O.K.**

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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4.2 Design of Retaining Wall around Ramp inside Concentrate Filtration Building

Retaining Wall of 1m Width & 2.5m high is modelled in Staad-Pro with relevant loads. It supports Ramp at a height of approximately 2m from base. Foundation Base of Retaining wall is at 1.0m below FGL.
There is compacted structural fill below 150 thk slab of Ramp. Supports are provided as plate mat with Modulus of subgrade reaction as 6000 T/m^2/m as per Geotechnical Report of BH-28

Load Calculation :

a) Dead Load (DL):

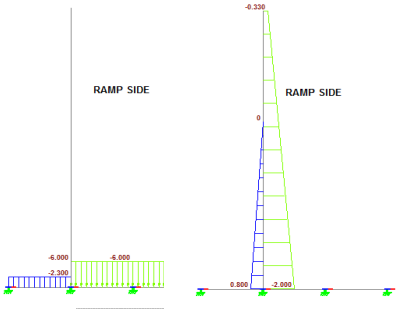
i) Self-Weight of Retaining Wall including Base Raft

ii) Wall Pressure due to compacted earthfill (applied as triangular distribution on wall in Staad)

As per dwg no. MD-216-3350-EG-ME-GAD-1053, Fork Lift Axle Load = 65 KN
Considering Axle wheel base = 0.85 For width of Ramp = 4 m
Surcharge on Ramp = 65/(4*0.85*10) = 2 T/m^2
Since the material stored is soil,
Angle of internal friction, ϕ considered = 30° ; k = (1-sinϕ)/(1+sinϕ) = 0.33
Self weight of Material Storage = 2* 2.5 5 T/m^2
Storage height considered from latest Tekla Model
Pressure on Heel Slab at wall base = 7 T/m^2
(excl. raft weight)
Pressure on Heel Slab at cantilever edge = 7 T/m^2
Pressure on Toe Slab = (1.5-0.3)*1.8 = 2.16 T/m^2

b) Horizontal Load due to material storage (H):

Lateral wall Pressure at top = 0.33*2 = 0.66 T/m^2
Lateral Wall Pressure at wall base
due to compacted earthfill on Ramp Side= 0.33 * (5 + 2) = 2.31 T/m^2
Lateral earth Pressure on other side = 2.16*0.33 = 0.7 T/m^2



DEAD LOAD (loads are in T/m) HORIZONTAL STORAGE LOAD (loads are in T/m)

c) Seismic Load (EL): (Calculated according to Chapter 7, SBC 303)

Seismic Parameters :

Seismic Design Category = B
Site Class = B
Seismic Coeffecient, Ss = 0.1 g (From Civil-Structural Design Criteria)
S_1 = 0.02 g
Occupancy importance factor, I = 1 (From Table 9.5, SBC 301:2007)
Response modification factor, R = 3 (From Table 10.2, SBC 301:2007)

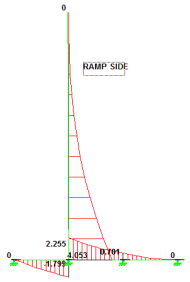
Height of the Wall, H = 2.5 m

Earthquake Load, Combined resultant Active Force, P = $1/2 * \gamma * H^2 * k_A * (1-k_v) * \cos^2\Theta / \cos^2\Theta \times \cos\zeta$ (eqn. 7-1, Cl. No. 7.2.6, SBC 303:2008)

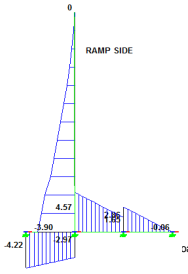
k_h = Horizontal Gound Acceleration = 0.1g =
k_v = Vertical Gound Acceleration = 0.1g *2/3 = 0.066g
ζ = $\tan^{-1}(k_h / 1-k_v)$ 0.107 (eqn. 7-2, Cl. No. 7.2.6, SBC 303:2008)
Θ = 0
$\Theta^* = \Theta + \zeta$ = ζ ; $\cos\zeta$ = 0.995

Combined resultant Active Force, P =
(0.5*(1.8*2.5+2)*2.5*0.33 *0.995*(1-0.067) = 2.49 T/m

Retaining wall has been designed for the following Load Combinations: (Table 5.3.1, SBC 304:2018) and ACI 318 : 2014
LOAD COMB 20 1.2*DL + 1.6*H
LOAD COMB 21 1.2DL+1.0EL
LOAD COMB 22 0.9DL+1.0EL + 1.6*H
Retaining wall has been designed for the following Bending Moment & Shear Force for LC- 0.9DL+1.0EL + 1.6*H (Most Critical):



BENDING MOMENT DIAGRAM



oads are in T-m)

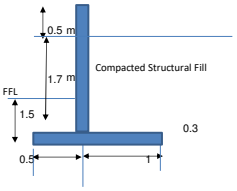
SHEAR FORCE DIAGRAM

Global Project ID	**900707**
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Stability Check for Retaining Wall



Lateral Wall Pressure at wall base = 2.4 T/m2
Width of Heel Slab from Center of Wall = 1 m

Combined resultant Active Force, P = 2.49 T/m
Overturning Moment = 2.49 * (0.6* 2.5) = 3.7 T-m
(Conservatively, Combined force acts at 0.6 times height of wall, as per Cl. No. 7.2.6, SBC 303)

Retaining Wall Sections :
Average Thickness of Wall = 300 mm
Thickness of Raft = 300 mm
Self weight of Retaining Wall = 2.8 T/m

Restoring Forces : (Passive Earth Pressure not considered for simplicity)
i) Weight of Stored Material = 1*1.7*2.5 = 4.25 T/m
ii) Weight of soil over Toe Slab = 1.08 T/m

Self Weight of Wall restoring moment = 2.775*0.5= 1.39 T-m (Restoring moment)

Total Weight = 8.1 T/m
Coeffecient of Friction between Concrete & Soil, μ = 0.4
Restoring Horizontal Force = 0.4*8.105 = 3.242 T/m

Moment due to Material = (1/2+0.5)*(1.7*1)*2.5 = 4.25 T-m/m
Total Restoring Moment = 5.64 T-m/m
For the combined effect of static & earthquake load ,
FOS against Overturning = 5.6375/3.735= 1.51 **> 1.2 - OK** (Refer Cl. No. 7.2.6, SBC 303)
FOS against Sliding = 3.242/2.49 = 1.30 **> 1.2 - OK** (Refer Cl. No. 7.2.6, SBC 303)

The Wall is Safe against Overturning & Sliding

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REINFORCEMENT CALCULATION OF WALL : (REFER STAAD OUTPUT FOR LC 20 TO 22)

Wall Thickness =	300	mm
Width =	1000	mm
Clear Cover =	50	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 1 T-m (ASSUMED)
Myd = 1 T-m

ALONG X - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1X10^7/(1000X232^2) = 0.186 N/mm^2 < deff. = 232 mm = 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X232/100 = 1.16 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0024

ALONG Y - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1X10^7/(1000X244^2) = 0.168 N/mm^2 < deff. = 244 mm = 6.9 N/mm^2
Pt = 0.0004 Reqd. Ast = 0.0004X1000X244/100 = 0.98 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Vertical
Total Provide Ast = 5.65 cm2 at Vertical **O.K.**
Total Provide Pt = 0.0023
Minimum Reinforcement ratio for both faces = 0.0025 (Cl. No. 11.6, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

Mxd = 1 T-M (ASSUMED)
Myd = 3.5 T-M (Refer Staad Output at face of raft) (LC -22 - 0.9DL+1.0EL + 1.6*H)

ALONG X - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1X10^7/(1000X232^2) = 0.186 N/mm^2 < deff. = 232 mm = 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X232/100 = 1.16 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0024

ALONG Y - DIRECTION :

Mu = 3.5 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 3.5X10^7/(X244^2) = 0.588 N/mm^2 < deff. = 244 mm = 6.9 N/mm^2
Pt = 0.0016 Reqd. Ast = 0.0016XX244/100 = 3.9 cm^2
Provided **12** mm @ **150** C/C (Provide Ast = 7.54 cm2) at Vertical

Total Provide Ast = 7.54 cm2) at Vertical **O.K.**
Total Provide Pt = 0.0031

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)
ALONG X - DIRECTION :

Vu = 4 T (Refer Staad Output at face of raftl)(LC - 22)

Provided Pt = 0.0031
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0031X4X10^4X244/(3.5/10^7)X1000X244/10^4
= 23.455 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0031)X1000X244/10^4
= 24.382 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X244/10^4
= 41.862 T
HENCE, Vc = 23.455 T
Φt*Vc = 0.75X23.455 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 17.591 T **> 4T , Hence O.K.**

Global Project ID	900707
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REINFORCEMENT CALCULATION OF WALL RAFT (REFER STAAD OUTPUT FOR LC 20 TO 22)

Slab Thickness =	300	mm
Width =	1000	mm
Clear Cover =	50	mm
Compressive Strength of Concrete , fc' =	35	N/mm^2
Yield Strength of Main Reinforcement , fy =	420	N/mm^2
Yield Strength of Shear Reinforcement , fyt =	420	N/mm^2
Strength Reduction Factor of Main Reinforcement , Φ =	0.9	
Strength Reduction Factor of Shear Reinforcement , Φt =	0.75	

DESIGN MOMENT CALCULATION FOR OPPOSITE FACE REINFORCEMENT :

Mxd = 2 T-m (Refer Staad Output at face of wall)(LC NO. 22)
Myd = 1 T-m (ASSUMED)

ALONG X - DIRECTION :

Mu = 2 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 2X10^7/(1000X244^2) = 0.336 N/mm^2 < deff. = 244 mm 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X244/100 = 2.2 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2 at Horizontal **O.K.**
Total Provide Pt = 0.0023

ALONG Y - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1X10^7/(1000X232^2) = 0.186 N/mm^2 < deff. = 232 mm 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X232/100 = 1.16 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0024
Minimum Reinforcement ratio for both faces = 0.0018 (Cl. No. Table 8.6.1.1, SBC 304 : 2018)

DESIGN MOMENT CALCULATION FOR RETAINED FACE REINFORCEMENT :

-0.036

Mxd = 2 T-M (Refer Staad Output at face of wall)(LC NO. 22)
Myd = 1 T-M (ASSUMED)

ALONG X - DIRECTION :

Mu = 2 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 2X10^7/(1000X244^2) = 0.336 N/mm^2 < deff. = 244 mm 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast = 0.0009X1000X244/100 = 2.2 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0023

ALONG Y - DIRECTION :

Mu = 1 T-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 1X10^7/(1000X232^2) = 0.186 N/mm^2 < deff. = 232 mm 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X232/100 = 1.16 cm^2
Provided **12** mm @ **200** C/C (Provide Ast = 5.65 cm2) at Horizontal
Total Provide Ast = 5.65 cm2) at Horizontal **O.K.**
Total Provide Pt = 0.0024

SHEAR CHECK
(FROM FACTORED LOAD COMBINATION)
ALONG X - DIRECTION :

Vu = 5 T (Refer Staad Output at a distance d from face of wall)(LC - 20 - 1.2DL+ 1.6*H)

Provided Pt = 0.0023
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0023X5X10^4X244/(2/10^7)X1000X244/10^4
= 23.678 T
[b] = (0.16*λ*fc^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0023)X1000X232/10^4
= 22.868 T
[c] = 0.29*λ*fc^0.5*b*d.eff
= 0.29X1X5.92X1000X232/10^4
= 39.803 T
HENCE, Vc = 22.868 T
Φt*Vc = 0.75X22.868 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 17.151 T > **5T , Hence O.K.**

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Project Country	**KINGDOM OF SAUDI ARABIA**

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REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

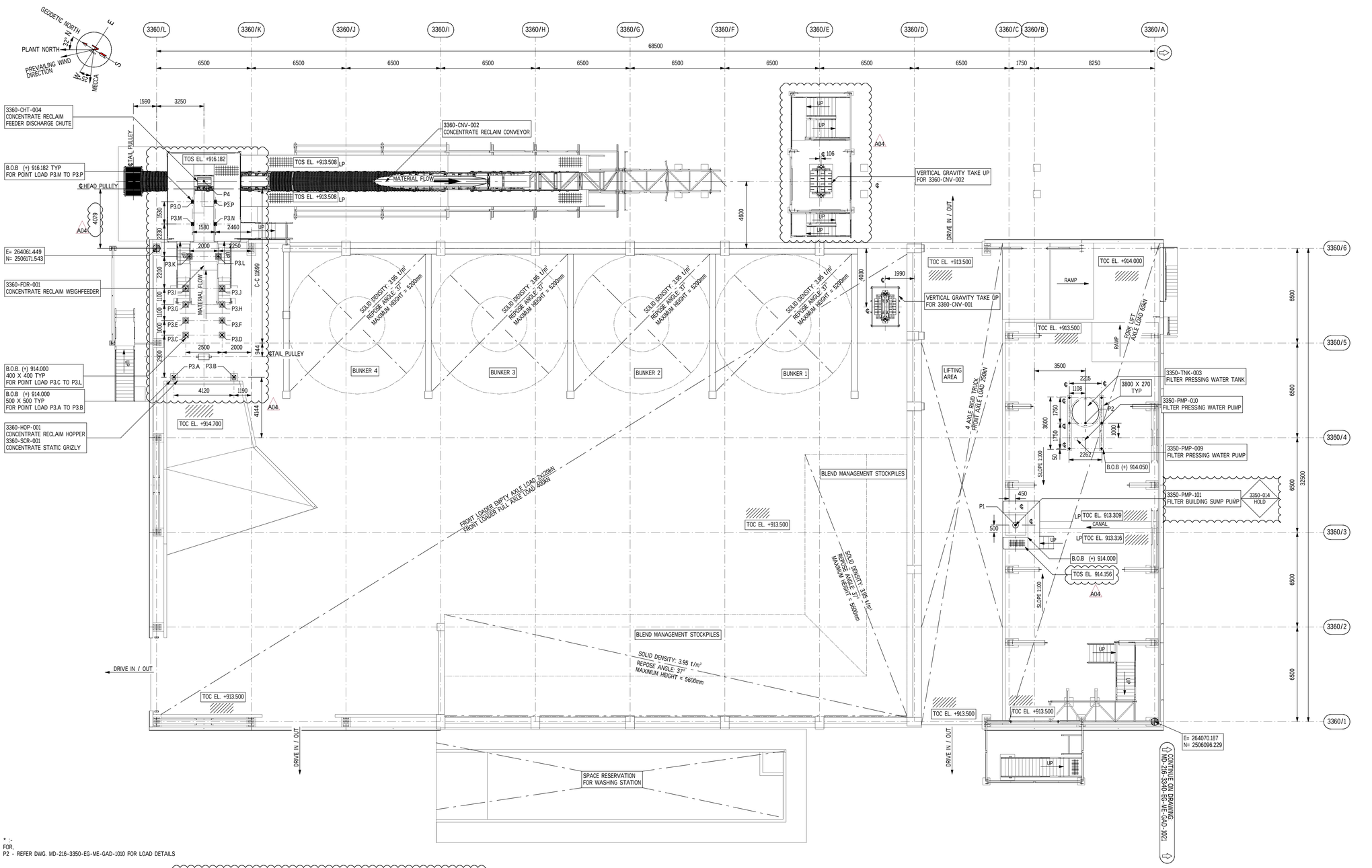
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01		
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI		

Outotec / L&T Revision No.:	0	1	2	3
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 1 CONCENTRATE FILTRATION LOAD PLAN MECHANICAL INPUT

DRAWING NUMBER: MD-216-3350-EG-ME-GAD-1053 | REV: A04



CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ALKALI / ACID RESISTANT COATING
CHECKERED PLATE
GRATING
GRATING - FRP
ACID RESISTANT BRICKLINING
BRICK WALL
CONCRETE WALL

GENERAL NOTES, LEGEND

01 - ALL DIMENSION ARE IN MILLIMETER. 02 - ALL COORDINATE AND LEVELS ARE IN METER.

03 - LOADS ARE INDICATED IN KILONEWTON[kN]; KILONEWTONMETER[kNm] OR KILONEWTON/ SQUAREMETER [kN/m²] UNLESS OTHERWISE NOTED.

04 - ALL LOADS REFER TO TOP OF BEAM UNLESS OTHERWISE NOTED.

05 - T.O.S., B.O.S, B.O.B. = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
T.O.C. = TOP OF CONCRETE
T.O.R. = TOP OF RAIL
LP = LOW POINT
HP = HIGHT POINT

DIRECTION OF FORCES AND MOMENTS

06 - THE LOADS SHOWN DO NOT INCLUDE :
- LOADS FROM AIR CONDITIONING EQUIPMENT
- ADDITIONAL LOADS FOR ERECTION PURPOSES

07 - FLOOR DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
OPERATING PLATFORM-GRATING 5 kN/m² GROUNDFLOOR NOT ACCESSIBLE TO TRUCKS 10 kN/m²
HEAVY STORAGE AREAS ACCESSIBLE BY VEHICLE OR CRANE 30 kN/m²
ELECTRICAL SWITCH GEAR AREAS 10 kN/m²

08 - ROOF DESIGNATION AND LIVE LOAD SHALL BE AS FOLLOW, UNLESS OTHERWISE NOTED:
DUST 0.5 kN/m² ROOF NOT ACCESSED FOR MAINTENANCE 0.8 kN POINT LOAD

09 - FOR ITEM SUCH AS DUCTS, PIPES, CABLE TRAYS, ETC. NOT SHOWN SEPARATELY, AN ADDITIONAL VERTICAL LOAD OF 50 kN SHALL BE PROVIDED ON TOP OF EACH COLUMN.

10 - STATIC LOADS :
FOR STATIC LOADS CERTAIN EQUIPMENT DATA ARE TAKEN INTO CONSIDERATION, AFTER PLACEMENT OF PURCHASE ORDER, STATIC LOADS FOR CERTAIN EQUIPMENT HAVE TO BE RECHECKED ACCORDING TO ACTUAL WEIGHTS.

11 - DYNAMIC LOADS :
INFORMATION WITH REGARD TO DYNAMIC LOADS WHICH ARE NOT INDICATED AND HAVE TO BE REQUESTED WITH THE SUPPLIER OF THE RESPECTIVE EQUIPMENT, THIS INFORMATION IS REQUIRED FOR THE DETAIL ENGINEERING AS A ROUGH ESTIMATION FOR THE DYNAMIC LOADS. TAKE 1.5 TIMES OF THE STATIC LOADS , BASED ON THE INDICATIONS FOR USUAL EQUIPMENT SUPPLIERS.

12 - REPORTED WIND LOADS ARE PRIMARY UNFACTORED LOADS SUITABLE FOR LOAD COMBINATIONS TO SBC301-2007. CIVIL DESIGNER TO APPLY COMBINATION FACTOR OF 1.6 FOR STRENGTH DESIGN.

13 - REPORTED SEISMIC LOADS ARE PRIMARY LOADS WITH A COMBINATION FACTOR OF 1.0 SUITABLE FOR STRENGTH DESIGN DERIVED IN ACCORDANCE WITH THE SITE SPECIFICATIONS AND THE REQUIREMENTS OF SBC301-2007. HOWEVER, IF THE LOADS ARE TO BE USED FOR ALLOWABLE STRESS DESIGN FOR FOUNDATION DESIGN THEN THE COMBINATION FACTOR WOULD BE 0.7 AND TO BE APPLIED BY CIVIL DESIGNER.

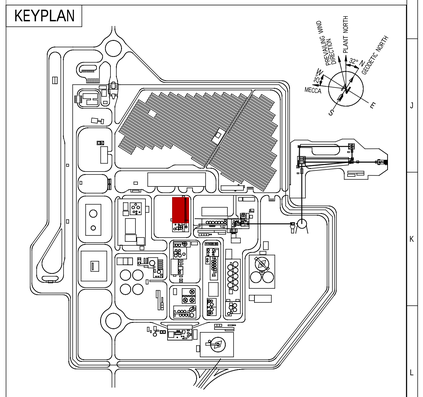
14 - FINISHED GROUND LEVEL (F.G.L.) = RL(+)913.000 m.

15 - LOCATIONS AND SIZES OF PUMP FOUNDATION ARE TENTATIVE AND WILL BE SHOWN IN DETAIL DESIGN PHASE.

16 - CONVEYOR LINE LOAD A = DEAD WEIGHT, B = MATERIAL WEIGHT.

17 - ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



* :-
FOR,
P2 - REFER DWG. MD-216-3350-EG-ME-GAD-1010 FOR LOAD DETAILS

LOADING POINT		P1			P2*			P3.A/B			P3.C/D			P3.E/F			P3.G/H			P3.I/J/K/L/M/N			P3.O/P			P4		
LOAD CASE		Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]	Vz[kN]	Hx[kN]	Hy[kN]
1	DEAD LOAD EQUIPMENT							- 12			- 40			- 16			- 101			- 30			- 48			- 3		
2	DEAD LOAD (SECONDARY EQUIPMENT)																											
3	OPERATION LOAD 1. FILLING																											
4	OPERATION LOAD 2. PRESSURE																											
5	OPERATION LOAD 3. TEMPERATURE																											
6	OPERATION LOAD 4. DYNAMIC	- 18																										
7	OPERATION LOAD 5. OTHERS																											
8	TEST LOAD FILLING / EMERGENCY																									- 85		
9	LIVE LOAD							- 92			- 396			- 152			- 82			- 66			- 72			- 50		
10	WIND ±X - DIRECTION																											
11	WIND ±Y - DIRECTION																											
12	EARTHQUAKE ±X - DIRECTION		0.6	-																						5	-	-
13	EARTHQUAKE ±Y - DIRECTION		-	0.6																						-	5	
	EQUIPMENT - NO.	3350-PMP-101			3350-TNK-003			3360-HOP-001			3360-HOP-001			3360-HOP-001			3360-HOP-001			3360-HOP-001			3360-HOP-001			3360-CHT-004		
	DESCRIPTION	FILTER BUILDING SUMP PUMP			3350-PMP-009, 3350-PMP-010			3360-SCR-001, 3360-FDR-001			3360-SCR-001, 3360-FDR-001			3360-SCR-001, 3360-FDR-001			3360-SCR-001, 3360-FDR-001			3360-SCR-001, 3360-FDR-001			3360-SCR-001, 3360-FDR-001			CONC. RECLAIM FEEDER DISH. CHUTE		

A04

PLAN AT EL.(+)913.500

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3350-EG-ME-GAD-1034	GENERAL ARRANGEMENT DRAWING, PLAN AT EL.+913.500
MD-216-3000-EG-GM-LST-1010	HOLD LIST
MD-216-3350-EG-ME-GAD-1056	LOAD PLAN, PLAN AT EL. +927.350
MD-216-3350-EG-ME-GAD-1055	LOAD PLAN, PLAN AT EL. +922.200 & EL. +923.900
MD-216-3350-EG-ME-GAD-1054	LOAD PLAN, PLAN VIEW

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A04	28-07-2020	REVISED AS MARKED	RW	TV	SC	RL	
A03	02-06-2020	REVISED AS MARKED	RW	TV	SC	RL	
A02	02-04-2020	ISSUED FOR DESIGN	RW	TV	SC	RL	
A01	28-01-2020	ISSUED FOR APPROVAL	BS	TV	SC	RL	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it or be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

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MA'ADEN معادن Saudi Arabian Mining Company

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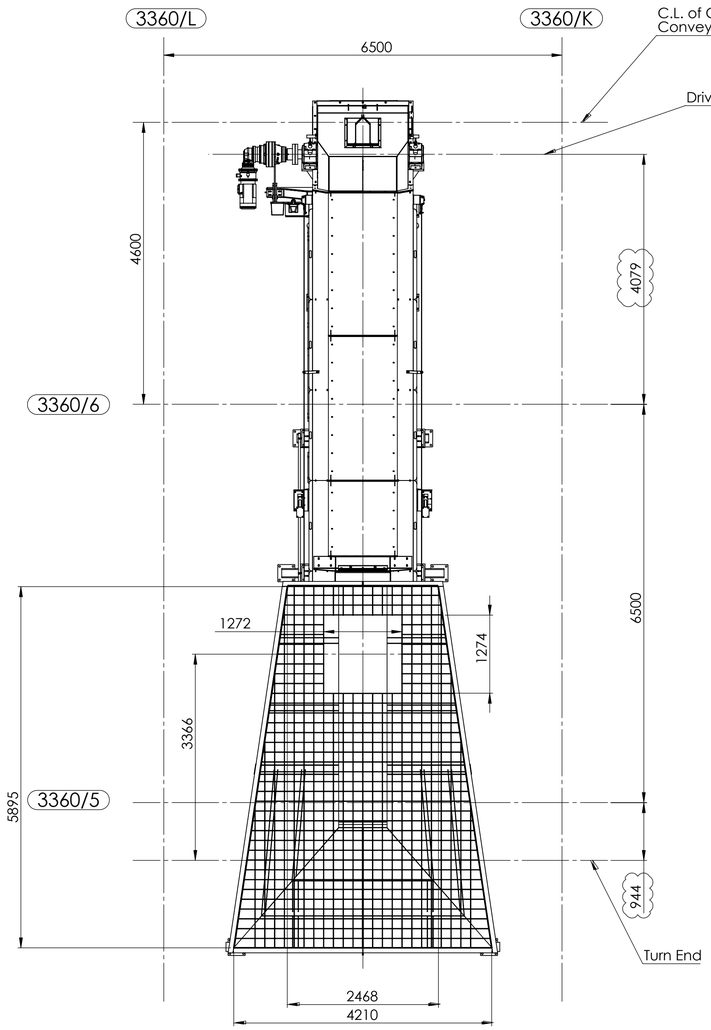
LARSEN & TOUBRO DRAWING NUMBER	019045_450_M_D_02468_001		REVISION 3
OUTOTEC DRAWING NUMBER	900707 _GAU01 _SF01 _MLD14 _33502	DOCUMENT ID 0U100667817	REVISION 3
DRAWN	BS	28-01-2020	MANSOURAH-MASSARAH GOLD PROJECT
CHECKED	TV	28-01-2020	LOAD PLAN
ENGINEER	SC	28-01-2020	CONCENTRATE FILTRATION
APPROVED	RL	28-01-2020	PLAN AT EL. +913.500
EMPLOYER			
SCALE 1:100	A0		DRAWING NUMBER MD-216-3350-EG-ME-GAD-1053 REV A04

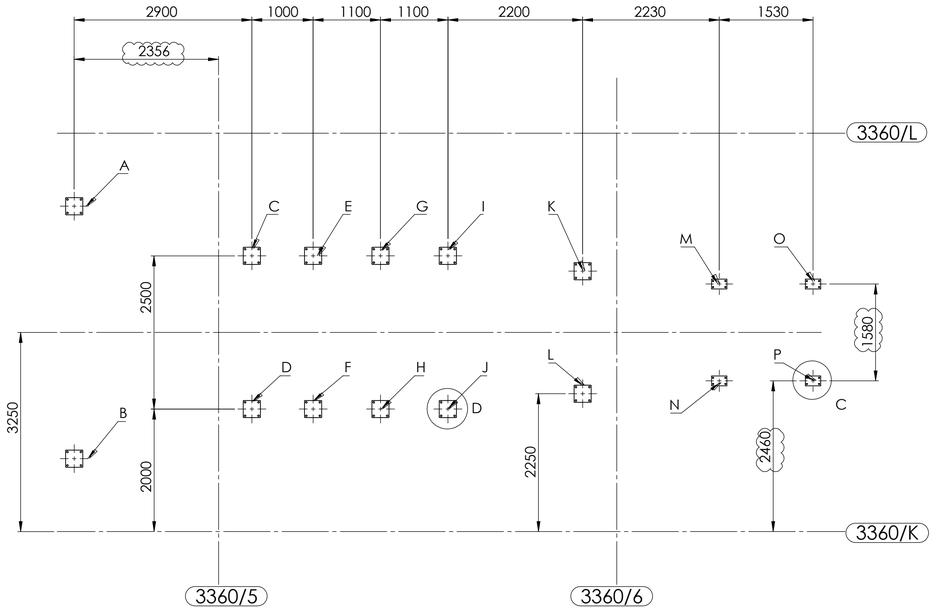
Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

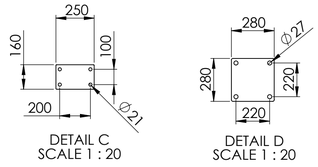
Ma'aden Revision No:	A01	A02	C01	D01
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED A02	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 2 LAYOUT & LOAD DATA OF CONCENTRATE RECLAIM WEIGHFEEDER

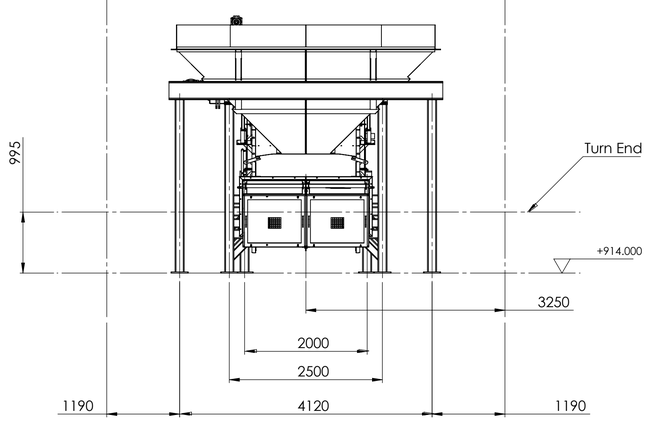


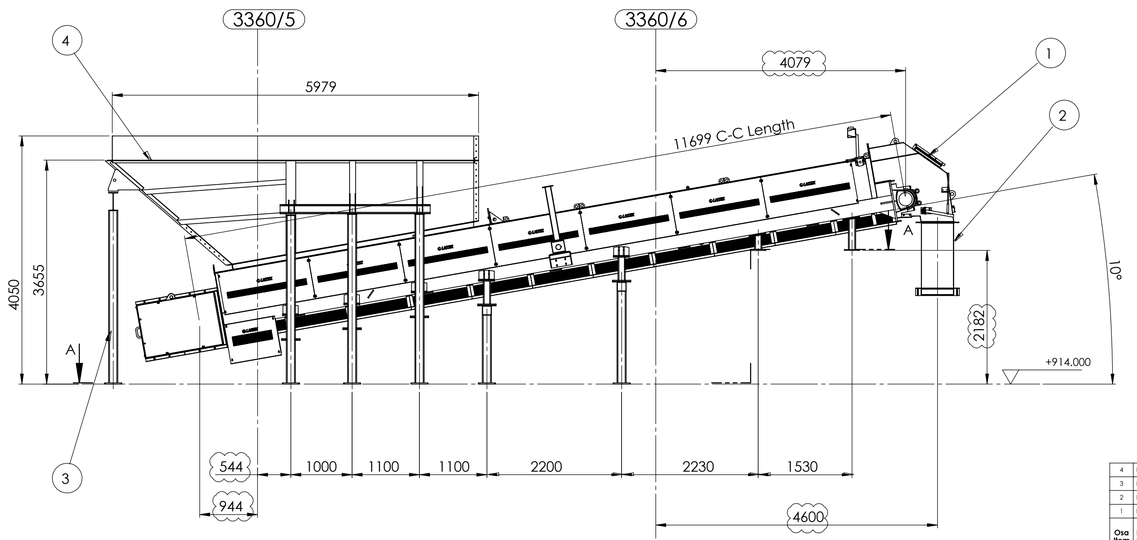


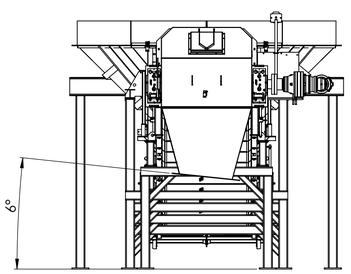


17.7.2020 For Approval

SECTION A-A







Item	Dwg no.	Description	Blank	Standard	Material	Dimensions	Notes	FC	SI	Mass / pcs	Qty
4	L1129793	HOPPER								3171	1
3	L1128287	STEEL STRUCTURE								1702	1
2	L1128151	CHUTE								184	1
1	L1125885	BELT CONVEYOR		LAITEX		HK 1200 x 11200	BCV-C 1200 x 11200			8407	1

Rev	Muutos / Change	Suunnittelija/Pvm. Designed/Date	Hyväksyjä/Pvm. Approved/Date
A	Euipment aligned correctly with next belt conveyor 3360-CNV-002	HL 17.07.2020	HL 17.07.2020

Asiakas: Outotec
Customer:
Työnumero: Laitetkoodi:
Work number: Device code:
Nimitys: LAYOUT
Description: LAYOUT
Kpl: 1
Pos:
Suunnittelija/Pvm: HL 04.06.2020
Designed/Date:
Hyväksyjä/Pvm: AM 10.07.2020
Approved/Date:
Massa: 13464kg
Mass:
Piirustusnumero: L1126176
Drawing number:
Revisio: A
Revision:
Mittakaava: 1:50
Scale:
Arkki: A1 1of2
Sheet:
Lisätietoja:
Add. information:
Projektipäällikkö/Project Manager:
LAITEX

Force reactions (kN)

Load case	Support A				Support B				Support C				Support D				Support E				Support F				Support G				Support H				Support I				Support J				Support K				Support L				Support M				Support N				Support O				Support P			
	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT
LC 1. Dead load 1x gravity	0	6	0	6	0	6	0	6	-6	19	0	20	6	19	0	20	-2	7	0	7	2	7	0	8	-1	5	0	5	1	5	0	5	-1	3	0	3	1	2	0	2	-1	5	0	5	0	5	0	5	-1	-4	-4	5	1	-3	-4	5	2	23	5	24	-1	14	4	14
LC 2. Live load 1.35xgravity 1.5x varibale loads	-5	45	1	46	3	45	1	45	-25	196	-3	198	24	189	-3	190	-5	75	-1	75	4	76	-1	76	4	38	-1	39	-4	41	-1	41	-3	11	0	11	3	11	0	11	-1	9	0	9	1	9	0	9	-1	-5	-9	11	1	-3	-9	9	3	35	6	36	-2	21	5	22
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
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	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Moment reactions (kNm)

Load case	Support A				Support B				Support C				Support D				Support E				Support F				Support G				Support H				Support I				Support J				Support K				Support L				Support M				Support N				Support O				Support P			
	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT	X	Y	Z	TOT
LC 1. Dead load 1x gravity	0	0	0	0	0	0	0	0	0	0	1	1	0	0	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-1	0	0	1	-1	0	0	1	1	0	0	1	1	0	0	1
LC 2. Live load 1.35xgravity 1.5x varibale loads	2	0	6	6	3	0	-2	3	0	-2	3	4	1	2	-2	3	1	-1	1	2	2	2	0	2	1	-1	-1	2	2	1	1	2	0	0	1	1	0	0	-1	1	0	0	0	0	0	0	0	0	-2	0	0	2	-1	0	0	1	1	0	0	1	1	0	0	1
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Load case	Load definition
LC1	Own mass with standard earth gravity (SEG)
LC2	Own mass with SEG*1,35 + 1,5*2,4tonne/m^3 material hydrostatic pressure on top of the hopper with SEG (max height 2385mm) + 1,5*material bed on belt (=138kg/m) with SEG

17.7.2020 For Approval

Rev	Muutos / Change	Suunnittelija/Pvm, Designed/Date	Hyväksyjä/Pvm, Approved/Date
A	Euipment aligned correctly with next belt conveyor 3360-CNV-002	HL 17.07.2020	HL 17.07.2020

up / down

levityskuvassa up ja down merkitsee painimen puolta/ up and down means upper knife side in flat pattern

Ruuviliitokset / Screw joints:

Ruuvien lujuusluokka: Strength class of screws:	8,8 (SFS-EN ISO 898-1)
Mutterien lujuusluokka: Strength class of nuts:	8,8 (SFS-EN ISO 898-1)
Kiristysmomentit: Tensioning moments:	M10 47 Nm; M12 81 Nm; M16 196 Nm; M20 385 Nm; M24 665 Nm; M30 1310 Nm

Kaikki hitsaukset SFS-EN ISO 5817 - C
Pienahitsit a-mitta 3,5 mm
Terävät särmät pyöristettävä

All welds according to SFS-EN ISO 5817 - C
Fillet welds a-dimension 3,5 mm
Sharp edges shall be rounded

(1) Kulman mittatoleranssi
(1) Tolerance of angle dimension
t = mm/m

Yleiset toleranssit: SFS-EN ISO 5817-C, SFS-EN ISO 9013-B, SFS-EN ISO 139208
General tolerances: SFS-EN ISO 13920-F, SFS-EN ISO 22768-1-mK

Nimellismitta: Nominal dimension	> 0 < 3	≥ 3 <10	≥ 10 < 35	≥35 < 125	≥ 125 < 315	≥ 315 < 1000	≥ 1000 < 2000	≥ 2000 < 4000	< 4000 < 8000	≥ 8000 < 12000	≥ 12000 < 16000	≥ 16000 < 20000	> 20000
Lastuava työstö: Machining: Etäisyys Distance	± 0,1	± 0,1	± 0,1	± 0,2	± 0,3	± 0,4	± 0,6	± 0,8	± 1,0	± 1,2	± 1,6	± 2	± 2,5
Leikkaus / taivutus: Cutting / bending:	± 0,2	± 0,2	± 0,5	± 0,8	± 1,2	± 2	± 3	± 4					
Terminen leikkaus; t = 3 - 50 mm Thermal cutting; t = 3 - 50 mm	-	-	-	± 1,8	± 1,9	± 2,3	± 3	± 4,2					
Hitsatut rakenteet (pituus): Welded constructions (linear):	± 1	± 1	± 1	± 2	± 2	± 3	± 4	± 6	± 8	± 10	± 12	± 14	± 16
Hitsatut rakenteet (kulma(1)): Welded constructions (angular(1)):	± 13	± 13	± 13	± 13	± 13	± 9	± 6	± 6					

Asiakas: Customer: Outotec

Työnumero: Work number:

Laitekoodi: Device code:

Nimitys: Description: LAYOUT LAYOUT

Kpl: Pcs: 1

Suunnittelija/Pvm: Designed/Date: HL 04.06.2020

Hyväksyjä/Pvm: Approved/Date: AM 10.07.2020

Massa: Mass: 13464kg

Piirustusnumero: Drawing number: L1126176

Revisio: Revision: A

Mittasuhde: Scale: 1:100

Arkki: Sheet: A3 2of2

Lisätietoja: Add. information:

Projektipäällikkö/Project Manager:

LAITEX conveying solutions

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01		
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI		

Outotec / L&T Revision No.:	0	1	2	3
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 3 STAAD Input & Output File of Foundation Mkd. F1

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG 29, 2020                *
*           Time=    13: 3:21                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: FDR-3360-001_FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 29-JUL-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 0 0 2.5; 3 1 0 0; 4 1 0 2.5; 5 2 0 0; 6 2 0 2.5; 7 3 0 0; 8 3 0 2.5
     9. 9 -0.25 0 0; 10 -0.25 0 2.5; 11 3.5 0 0; 12 3.5 0 2.5; 13 0 0 -0.25
    10. 14 1 0 -0.25; 15 2 0 -0.25; 16 3 0 -0.25; 17 -0.25 0 -0.25; 18 3.5 0 -0.25
    11. 19 0 0 2.75; 20 1 0 2.75; 21 2 0 2.75; 22 3 0 2.75; 23 -0.25 0 2.75
    12. 24 3.5 0 2.75; 25 0.25 0 0; 26 0.25 0 -0.25; 27 0.5 0 0; 28 0.5 0 -0.25
    13. 29 0.75 0 0; 30 0.75 0 -0.25; 31 1.25 0 0; 32 1.25 0 -0.25; 33 1.5 0 0
    14. 34 1.5 0 -0.25; 35 1.75 0 0; 36 1.75 0 -0.25; 37 2.25 0 0; 38 2.25 0 -0.25
    15. 39 2.5 0 0; 40 2.5 0 -0.25; 41 2.75 0 0; 42 2.75 0 -0.25; 43 3.25 0 0
    16. 44 3.25 0 -0.25; 47 0.25 0 2.75; 48 0.25 0 2.5; 49 0.5 0 2.75; 50 0.5 0 2.5
    17. 51 0.75 0 2.75; 52 0.75 0 2.5; 53 1.25 0 2.75; 54 1.25 0 2.5; 55 1.5 0 2.75
    18. 56 1.5 0 2.5; 57 1.75 0 2.75; 58 1.75 0 2.5; 59 2.25 0 2.75; 60 2.25 0 2.5
    19. 61 2.5 0 2.75; 62 2.5 0 2.5; 63 2.75 0 2.75; 64 2.75 0 2.5; 65 3.25 0 2.75
    20. 66 3.25 0 2.5; 69 0 0 0.25; 70 -0.25 0 0.25; 71 0 0 0.5; 72 -0.25 0 0.5
    21. 73 0 0 0.75; 74 -0.25 0 0.75; 75 0 0 1; 76 -0.25 0 1; 77 0 0 1.25
    22. 78 -0.25 0 1.25; 79 0 0 1.5; 80 -0.25 0 1.5; 81 0 0 1.75; 82 -0.25 0 1.75
    23. 83 0 0 2; 84 -0.25 0 2; 85 0 0 2.25; 86 -0.25 0 2.25; 87 1 0 0.25; 88 2 0 0.25
    24. 89 3 0 0.25; 90 3.5 0 0.25; 91 0.25 0 0.25; 92 0.5 0 0.25; 93 0.75 0 0.25
    25. 94 1.25 0 0.25; 95 1.5 0 0.25; 96 1.75 0 0.25; 97 2.25 0 0.25; 98 2.5 0 0.25
    26. 99 2.75 0 0.25; 100 3.25 0 0.25; 102 1 0 0.5; 103 2 0 0.5; 104 3 0 0.5
    27. 105 3.5 0 0.5; 106 0.25 0 0.5; 107 0.5 0 0.5; 108 0.75 0 0.5; 109 1.25 0 0.5
    28. 110 1.5 0 0.5; 111 1.75 0 0.5; 112 2.25 0 0.5; 113 2.5 0 0.5; 114 2.75 0 0.5
    29. 115 3.25 0 0.5; 117 1 0 0.75; 118 2 0 0.75; 119 3 0 0.75; 120 3.5 0 0.75
    30. 121 0.25 0 0.75; 122 0.5 0 0.75; 123 0.75 0 0.75; 124 1.25 0 0.75
    31. 125 1.5 0 0.75; 126 1.75 0 0.75; 127 2.25 0 0.75; 128 2.5 0 0.75
    32. 129 2.75 0 0.75; 130 3.25 0 0.75; 132 1 0 1; 133 2 0 1; 134 3 0 1; 135 3.5 0 1
    33. 136 0.25 0 1; 137 0.5 0 1; 138 0.75 0 1; 139 1.25 0 1; 140 1.5 0 1
    34. 141 1.75 0 1; 142 2.25 0 1; 143 2.5 0 1; 144 2.75 0 1; 145 3.25 0 1
    35. 147 1 0 1.25; 148 2 0 1.25; 149 3 0 1.25; 150 3.5 0 1.25; 151 0.25 0 1.25
    36. 152 0.5 0 1.25; 153 0.75 0 1.25; 154 1.25 0 1.25; 155 1.5 0 1.25
    37. 156 1.75 0 1.25; 157 2.25 0 1.25; 158 2.5 0 1.25; 159 2.75 0 1.25
    38. 160 3.25 0 1.25; 162 1 0 1.5; 163 2 0 1.5; 164 3 0 1.5; 165 3.5 0 1.5
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39. 166 0.25 0 1.5; 167 0.5 0 1.5; 168 0.75 0 1.5; 169 1.25 0 1.5; 170 1.5 0 1.5
40. 171 1.75 0 1.5; 172 2.25 0 1.5; 173 2.5 0 1.5; 174 2.75 0 1.5; 175 3.25 0 1.5
41. 177 1 0 1.75; 178 2 0 1.75; 179 3 0 1.75; 180 3.5 0 1.75; 181 0.25 0 1.75
42. 182 0.5 0 1.75; 183 0.75 0 1.75; 184 1.25 0 1.75; 185 1.5 0 1.75
43. 186 1.75 0 1.75; 187 2.25 0 1.75; 188 2.5 0 1.75; 189 2.75 0 1.75
44. 190 3.25 0 1.75; 192 1 0 2; 193 2 0 2; 194 3 0 2; 195 3.5 0 2; 196 0.25 0 2
45. 197 0.5 0 2; 198 0.75 0 2; 199 1.25 0 2; 200 1.5 0 2; 201 1.75 0 2
46. 202 2.25 0 2; 203 2.5 0 2; 204 2.75 0 2; 205 3.25 0 2; 207 1 0 2.25
47. 208 2 0 2.25; 209 3 0 2.25; 210 3.5 0 2.25; 211 0.25 0 2.25; 212 0.5 0 2.25
48. 213 0.75 0 2.25; 214 1.25 0 2.25; 215 1.5 0 2.25; 216 1.75 0 2.25
49. 217 2.25 0 2.25; 218 2.5 0 2.25; 219 2.75 0 2.25; 220 3.25 0 2.25; 222 0 2.5 0
50. 223 0 2.5 2.5; 224 1 2.5 0; 225 1 2.5 2.5; 226 2 2.5 0; 227 2 2.5 2.5
51. 228 3 2.5 0; 229 3 2.5 2.5; 230 -0.25 0 -0.5; 231 0 0 -0.5; 232 0.25 0 -0.5
52. 233 0.5 0 -0.5; 234 0.75 0 -0.5; 235 1 0 -0.5; 236 1.25 0 -0.5; 237 1.5 0 -0.5
53. 238 1.75 0 -0.5; 239 2 0 -0.5; 240 2.25 0 -0.5; 241 2.5 0 -0.5
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55. 247 0 0 3; 248 0.25 0 3; 249 0.5 0 3; 250 0.75 0 3; 251 1 0 3; 252 1.25 0 3
56. 253 1.5 0 3; 254 1.75 0 3; 255 2 0 3; 256 2.25 0 3; 257 2.5 0 3; 258 2.75 0 3
57. 259 3 0 3; 260 3.25 0 3; 261 3.5 0 3; 262 -0.5 0 -0.25; 263 -0.5 0 0
58. 264 -0.5 0 2.5; 265 -0.5 0 2.75; 266 -0.5 0 0.25; 267 -0.5 0 0.5
59. 268 -0.5 0 0.75; 269 -0.5 0 1; 270 -0.5 0 1.25; 271 -0.5 0 1.5
60. 272 -0.5 0 1.75; 273 -0.5 0 2; 274 -0.5 0 2.25; 275 -0.5 0 -0.5; 276 -0.5 0 3
61. MEMBER INCIDENCES
62. 193 1 222; 194 2 223; 195 3 224; 196 4 225; 197 5 226; 198 6 227; 199 7 228
63. 200 8 229
64. ELEMENT INCIDENCES SHELL
65. 1 17 9 1 13; 2 13 1 25 26; 3 26 25 27 28; 4 28 27 29 30; 5 30 29 3 14
66. 6 14 3 31 32; 7 32 31 33 34; 8 34 33 35 36; 9 36 35 5 15; 10 15 5 37 38
67. 11 38 37 39 40; 12 40 39 41 42; 13 42 41 7 16; 14 16 7 43 44; 15 44 43 11 18
68. 17 10 23 19 2; 18 2 19 47 48; 19 48 47 49 50; 20 50 49 51 52; 21 52 51 20 4
69. 22 4 20 53 54; 23 54 53 55 56; 24 56 55 57 58; 25 58 57 21 6; 26 6 21 59 60
70. 27 60 59 61 62; 28 62 61 63 64; 29 64 63 22 8; 30 8 22 65 66; 31 66 65 24 12
71. 33 9 70 69 1; 34 70 72 71 69; 35 72 74 73 71; 36 74 76 75 73; 37 76 78 77 75
72. 38 78 80 79 77; 39 80 82 81 79; 40 82 84 83 81; 41 84 86 85 83; 42 86 10 2 85
73. 43 1 69 91 25; 44 25 91 92 27; 45 27 92 93 29; 46 29 93 87 3; 47 3 87 94 31
74. 48 31 94 95 33; 49 33 95 96 35; 50 35 96 88 5; 51 5 88 97 37; 52 37 97 98 39
75. 53 39 98 99 41; 54 41 99 89 7; 55 7 89 100 43; 56 43 100 90 11
76. 58 69 71 106 91; 59 91 106 107 92; 60 92 107 108 93; 61 93 108 102 87
77. 62 87 102 109 94; 63 94 109 110 95; 64 95 110 111 96; 65 96 111 103 88
78. 66 88 103 112 97; 67 97 112 113 98; 68 98 113 114 99; 69 99 114 104 89
79. 70 89 104 115 100; 71 100 115 105 90; 73 71 73 121 106; 74 106 121 122 107
80. 75 107 122 123 108; 76 108 123 117 102; 77 102 117 124 109; 78 109 124 125 110
81. 79 110 125 126 111; 80 111 126 118 103; 81 103 118 127 112; 82 112 127 128 113
82. 83 113 128 129 114; 84 114 129 119 104; 85 104 119 130 115; 86 115 130 120 105
83. 88 73 75 136 121; 89 121 136 137 122; 90 122 137 138 123; 91 123 138 132 117
84. 92 117 132 139 124; 93 124 139 140 125; 94 125 140 141 126; 95 126 141 133 118
85. 96 118 133 142 127; 97 127 142 143 128; 98 128 143 144 129; 99 129 144 134 119
86. 100 119 134 145 130; 101 130 145 135 120; 103 75 77 151 136
87. 104 136 151 152 137; 105 137 152 153 138; 106 138 153 147 132
88. 107 132 147 154 139; 108 139 154 155 140; 109 140 155 156 141
89. 110 141 156 148 133; 111 133 148 157 142; 112 142 157 158 143
90. 113 143 158 159 144; 114 144 159 149 134; 115 134 149 160 145
91. 116 145 160 150 135; 118 77 79 166 151; 119 151 166 167 152
92. 120 152 167 168 153; 121 153 168 162 147; 122 147 162 169 154
93. 123 154 169 170 155; 124 155 170 171 156; 125 156 171 163 148
94. 126 148 163 172 157; 127 157 172 173 158; 128 158 173 174 159
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 95. 129 159 174 164 149; 130 149 164 175 160; 131 160 175 165 150
 96. 133 79 81 181 166; 134 166 181 182 167; 135 167 182 183 168
 97. 136 168 183 177 162; 137 162 177 184 169; 138 169 184 185 170
 98. 139 170 185 186 171; 140 171 186 178 163; 141 163 178 187 172
 99. 142 172 187 188 173; 143 173 188 189 174; 144 174 189 179 164
100. 145 164 179 190 175; 146 175 190 180 165; 148 81 83 196 181
101. 149 181 196 197 182; 150 182 197 198 183; 151 183 198 192 177
102. 152 177 192 199 184; 153 184 199 200 185; 154 185 200 201 186
103. 155 186 201 193 178; 156 178 193 202 187; 157 187 202 203 188
104. 158 188 203 204 189; 159 189 204 194 179; 160 179 194 205 190
105. 161 190 205 195 180; 163 83 85 211 196; 164 196 211 212 197
106. 165 197 212 213 198; 166 198 213 207 192; 167 192 207 214 199
107. 168 199 214 215 200; 169 200 215 216 201; 170 201 216 208 193
108. 171 193 208 217 202; 172 202 217 218 203; 173 203 218 219 204
109. 174 204 219 209 194; 175 194 209 220 205; 176 205 220 210 195; 178 85 2 48 211
110. 179 211 48 50 212; 180 212 50 52 213; 181 213 52 4 207; 182 207 4 54 214
111. 183 214 54 56 215; 184 215 56 58 216; 185 216 58 6 208; 186 208 6 60 217
112. 187 217 60 62 218; 188 218 62 64 219; 189 219 64 8 209; 190 209 8 66 220
113. 191 220 66 12 210; 201 230 17 13 231; 202 231 13 26 232; 203 232 26 28 233
114. 204 233 28 30 234; 205 234 30 14 235; 206 235 14 32 236; 207 236 32 34 237
115. 208 237 34 36 238; 209 238 36 15 239; 210 239 15 38 240; 211 240 38 40 241
116. 212 241 40 42 242; 213 242 42 16 243; 214 243 16 44 244; 215 244 44 18 245
117. 216 23 246 247 19; 217 19 247 248 47; 218 47 248 249 49; 219 49 249 250 51
118. 220 51 250 251 20; 221 20 251 252 53; 222 53 252 253 55; 223 55 253 254 57
119. 224 57 254 255 21; 225 21 255 256 59; 226 59 256 257 61; 227 61 257 258 63
120. 228 63 258 259 22; 229 22 259 260 65; 230 65 260 261 24; 231 262 263 9 17
121. 232 264 265 23 10; 233 263 266 70 9; 234 266 267 72 70; 235 267 268 74 72
122. 236 268 269 76 74; 237 269 270 78 76; 238 270 271 80 78; 239 271 272 82 80
123. 240 272 273 84 82; 241 273 274 86 84; 242 274 264 10 86; 243 275 262 17 230
124. 244 265 276 246 23
125. START GROUP DEFINITION
126. ELEMENT
127. _MAX_BM 3 4 7 8 11 12 15 19 20 23 24 27 28 31 34 TO 41 44 45 48 49 52 53 56 -
128. 58 TO 71 73 TO 86 88 TO 101 103 TO 116 118 TO 131 133 TO 146 148 TO 161 163 -
129. 164 TO 176 179 180 183 184 187 188 191 201 TO 244
130. _MAX_SF 35 TO 40 73 TO 86 88 TO 101 103 TO 116 118 TO 131 133 TO 146 -
131. 148 TO 161 235 TO 240
132. END GROUP DEFINITION
133. ELEMENT PROPERTY
134. 1 TO 15 17 TO 31 33 TO 56 58 TO 71 73 TO 86 88 TO 101 103 TO 116 118 TO 131 -
135. 133 TO 146 148 TO 161 163 TO 176 178 TO 191 201 TO 244 THICKNESS 0.5
136. DEFINE MATERIAL START
137. ISOTROPIC CONCRETE
138. E 2.21467E+006
139. POISSON 0.17
140. DENSITY 2.40262
141. ALPHA 1E-005
142. DAMP 0.05
143. TYPE CONCRETE
144. STRENGTH FCU 2812.28
145. END DEFINE MATERIAL
146. MEMBER PROPERTY AMERICAN
147. 193 TO 200 PRIS YD 0.45 ZD 0.45
148. CONSTANTS
149. MATERIAL CONCRETE ALL
150. SUPPORTS
```

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151. 1 TO 15 17 TO 31 33 34 36 TO 39 41 TO 56 58 TO 71 88 TO 101 103 TO 116 118 -
152. 119 TO 131 133 TO 146 163 TO 176 178 TO 191 201 TO 234 236 TO 239 241 TO 243 -
153. 244 PLATE MAT DIRECT Y SUBGRADE 5400 PRINT COMPRESSION
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
3	0.062500
4	0.062500
5	0.062500
6	0.062500
7	0.062500
8	0.062500
9	0.062500
10	0.062500
11	0.031250
12	0.031250
13	0.062500
14	0.062500
15	0.062500
16	0.062500
17	0.062500
18	0.031250
19	0.062500
20	0.062500
21	0.062500
22	0.062500
23	0.062500
24	0.031250
25	0.062500
26	0.062500
27	0.062500
28	0.062500
29	0.062500
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.062500
36	0.062500
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38	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062500
47	0.062500
48	0.062500
49	0.062500
50	0.062500
51	0.062500

52	0.062500
53	0.062500
54	0.062500
55	0.062500
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58	0.062500
59	0.062500
60	0.062500
61	0.062500
62	0.062500
63	0.062500
64	0.062500
65	0.062500
66	0.062500
69	0.062500
70	0.062500
71	0.031250
72	0.031250
73	0.031250
74	0.031250
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062500
80	0.062500
81	0.031250
82	0.031250
83	0.031250
84	0.031250
85	0.062500
86	0.062500
87	0.062500
88	0.062500
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93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062500
98	0.062500
99	0.062500
100	0.062500
102	0.031250
103	0.031250
104	0.031250
105	0.015625
106	0.031250
107	0.031250
108	0.031250
109	0.031250
110	0.031250

111	0.031250
112	0.031250
113	0.031250
114	0.031250
115	0.031250
117	0.031250
118	0.031250
119	0.031250
120	0.015625
121	0.031250
122	0.031250
123	0.031250
124	0.031250
125	0.031250
126	0.031250
127	0.031250
128	0.031250
129	0.031250
130	0.031250
132	0.062500
133	0.062500
134	0.062500
135	0.031250
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062500
142	0.062500
143	0.062500
144	0.062500
145	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.031250
151	0.062500
152	0.062500
153	0.062500
154	0.062500
155	0.062500
156	0.062500
157	0.062500
158	0.062500
159	0.062500
160	0.062500
162	0.062500
163	0.062500
164	0.062500
165	0.031250
166	0.062500
167	0.062500
168	0.062500
169	0.062500
170	0.062500

171	0.062500
172	0.062500
173	0.062500
174	0.062500
175	0.062500
177	0.031250
178	0.031250
179	0.031250
180	0.015625
181	0.031250
182	0.031250
183	0.031250
184	0.031250
185	0.031250
186	0.031250
187	0.031250
188	0.031250
189	0.031250
190	0.031250
192	0.031250
193	0.031250
194	0.031250
195	0.015625
196	0.031250
197	0.031250
198	0.031250
199	0.031250
200	0.031250
201	0.031250
202	0.031250
203	0.031250
204	0.031250
205	0.031250
207	0.062500
208	0.062500
209	0.062500
210	0.031250
211	0.062500
212	0.062500
213	0.062500
214	0.062500
215	0.062500
216	0.062500
217	0.062500
218	0.062500
219	0.062500
220	0.062500
230	0.031250
231	0.031250
232	0.031250
233	0.031250
234	0.031250
235	0.031250
236	0.031250
237	0.031250
238	0.031250

239	0.031250
240	0.031250
241	0.031250
242	0.031250
243	0.031250
244	0.031250
245	0.015625
246	0.031250
247	0.031250
248	0.031250
249	0.031250
250	0.031250
251	0.031250
252	0.031250
253	0.031250
254	0.031250
255	0.031250
256	0.031250
257	0.031250
258	0.031250
259	0.031250
260	0.031250
261	0.015625
262	0.031250
263	0.031250
264	0.031250
265	0.031250
266	0.031250
267	0.015625
268	0.015625
269	0.031250
270	0.031250
271	0.031250
272	0.015625
273	0.015625
274	0.031250
275	0.015625
276	0.015625
TOTAL	12.000000

```
154. LOAD 1 LOADTYPE NONE  TITLE DL
155. JOINT LOAD
156. 222 223 FY -2
157. 224 225 FY -0.7
158. 226 227 FY -0.5
159. 228 229 FY -0.3
160. ELEMENT LOAD
161. 1 TO 15 17 TO 31 33 34 36 TO 39 41 TO 56 58 TO 71 88 TO 101 103 TO 116 118 -
162. 119 TO 131 133 TO 146 163 TO 176 178 TO 191 201 TO 234 236 TO 239 241 TO 243 -
163. 244 PR GY -4
164. SELFWEIGHT Y -1
165. LOAD 2 LOADTYPE NONE  TITLE LL
166. JOINT LOAD
167. 222 FX 2.5 FY -20 FZ 0.3 MX 0.1 MZ 0.1
168. 223 FX -2.5 FY -20 FZ 0.3 MX 0.1 MZ 0.1
169. 224 FX 0.5 FY -7.5 FZ 0.2 MX 0.1 MZ 0.1
170. 225 FX -0.5 FY -7.5 FZ 0.2 MX 0.1 MZ 0.1
```

```
171. 226 FX 0.4 FY -4 FZ 0.1 MX 0.1 MZ 0.1
172. 227 FX -0.4 FY -4 FZ 0.1 MX 0.1 MZ 0.1
173. 228 FX 0.3 FY -1.1 FZ 0.1 MX 0.1 MZ 0.1
174. 229 FX -0.3 FY -1.1 FZ 0.1 MX 0.1 MZ 0.1
175. *WORKING STRESS LOAD COMB*
176. LOAD COMB 10 DL+LL
177. 1 1.0 2 1.0
178. *ULTIMATE STREGTH LOAD COMB*
179. LOAD COMB 20 1.4DL+1.7LL
180. 1 1.4 2 1.7
181. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**263**	**NUMBER OF MEMBERS**	**8**
NUMBER OF PLATES	**224**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**255**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 251/ 26/ 87 DOF
TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 813
TOTAL LOAD COMBINATION CASES = 2 SO FAR.
SIZE OF STIFFNESS MATRIX = 71 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.0/ 163625.6 MB

```
**NOTE-Tension/Compression converged after  1 iterations, Case=    1


**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4

**NOTE-Tension/Compression converged after  4 iterations, Case=    2


182. START CONCRETE DESIGN
```

```
183. CODE ACI 1999
184. TRACK 0 ALL
185. MAXMAIN 25 ALL
186. CLB 0.075 MEMB 193 TO 200
187. CLS 0.075 ALL
188. FC 3058 MEMB 193 TO 200
189. DESIGN COLUMN 193 TO 200
```

==

COLUMN NO. 193 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 194 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 195 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 196 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 197 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 198 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 199 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==========

COLUMN NO. 200 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	1	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

```
190. END CONCRETE DESIGN
191. FINISH


     ********** END OF THE STAAD.Pro RUN **********

       **** DATE= AUG 29,2020   TIME= 13: 3:35 ****

   ************************************************************
   *   For technical assistance on STAAD.Pro, please visit    *
   *   http://selectservices.bentley.com/en-US/               *
   *                                                          *
   *   Details about additional assistance from               *
   *   Bentley and Partners can be found at program menu      *
   *   Help->Technical Support                                *
   *                                                          *
   *      Copyright (c) 1997-2014 Bentley Systems, Inc.
   *                http://www.bentley.com                    *
   ************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 4 STAAD Input & Output File of Foundation Mkd. F2

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    SEP  4, 2020                *
*           Time=    17:23: 8                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: C1-C2_FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 -1.25; 14 0 -4 -1.5; 16 0 -4 -0.25; 24 0.5 -4 -0.25; 25 0.5 -4 -0.5
     9. 26 0.25 -4 -0.25; 27 0.25 -4 -0.5; 28 0 -4 -0.5; 31 0.5 -4 -0.75
    10. 32 0.25 -4 -0.75; 33 0 -4 -0.75; 36 0.5 -4 -1; 37 0.25 -4 -1; 38 0 -4 -1.
    11. 41 0.5 -4 -1.25; 42 0.25 -4 -1.25; 70 0.5 -4 -1.5; 71 0.25 -4 -1.5
    12. 74 0.5 -4 -1.75; 75 0.25 -4 -1.75; 76 0 -4 -1.75; 79 0.5 -4 -2; 80 0.25 -4 -2
    13. 81 0 -4 -2; 84 0.5 -4 -2.25; 85 0.25 -4 -2.25; 86 0 -4 -2.25
    14. 90 -0.25 -4 -0.25; 91 -0.25 -4 -0.5; 92 -0.5 -4 -0.25; 93 -0.5 -4 -0.5
    15. 97 -0.25 -4 -0.75; 98 -0.5 -4 -0.75; 101 -0.25 -4 -1; 102 -0.5 -4 -1.
    16. 105 -0.25 -4 -1.25; 106 -0.5 -4 -1.25; 129 -0.25 -4 -1.5; 130 -0.5 -4 -1.5
    17. 133 -0.25 -4 -1.75; 134 -0.5 -4 -1.75; 137 -0.25 -4 -2; 138 -0.5 -4 -2
    18. 141 -0.25 -4 -2.25; 142 -0.5 -4 -2.25; 145 0 -1.8 -2; 146 0 -1.8 -0.5
    19. 147 -0.6 -4 -0.25; 148 -0.6 -4 -0.5; 149 -0.6 -4 -0.75; 150 -0.6 -4 -1.
    20. 151 -0.6 -4 -1.25; 152 -0.6 -4 -1.5; 153 -0.6 -4 -1.75; 154 -0.6 -4 -2
    21. 155 -0.6 -4 -2.25; 156 0.6 -4 -0.25; 157 0.6 -4 -0.5; 158 0.6 -4 -0.75
    22. 159 0.6 -4 -1; 160 0.6 -4 -1.25; 161 0.6 -4 -1.5; 162 0.6 -4 -1.75
    23. 163 0.6 -4 -2; 164 0.6 -4 -2.25; 165 0.25 -4 -2.5; 166 0.5 -4 -2.5
    24. 167 0 -4 -2.5; 168 -0.25 -4 -2.5; 169 -0.5 -4 -2.5; 170 -0.6 -4 -2.5
    25. 171 0.6 -4 -2.5; 172 0.5 -4 0; 173 0.25 -4 0; 174 0 -4 0; 175 -0.25 -4 0
    26. 176 -0.5 -4 0; 177 -0.6 -4 0; 178 0.6 -4 0
    27. MEMBER INCIDENCES
    28. 133 81 145; 134 28 146
    29. ELEMENT INCIDENCES SHELL
    30. 27 24 26 27 25; 28 26 16 28 27; 31 25 27 32 31; 32 27 28 33 32; 35 31 32 37 36
    31. 36 32 33 38 37; 39 36 37 42 41; 40 37 38 13 42; 63 41 42 71 70; 64 42 13 14 71
    32. 67 70 71 75 74; 68 71 14 76 75; 71 74 75 80 79; 72 75 76 81 80; 75 79 80 85 84
    33. 76 80 81 86 85; 81 16 90 91 28; 82 90 92 93 91; 85 28 91 97 33; 86 91 93 98 97
    34. 89 33 97 101 38; 90 97 98 102 101; 93 38 101 105 13; 94 101 102 106 105
    35. 117 13 105 129 14; 118 105 106 130 129; 121 14 129 133 76; 122 129 130 134 133
    36. 125 76 133 137 81; 126 133 134 138 137; 129 81 137 141 86; 130 137 138 142 141
    37. 135 92 147 148 93; 136 93 148 149 98; 137 98 149 150 102; 138 102 150 151 106
    38. 139 106 151 152 130; 140 130 152 153 134; 141 134 153 154 138
```

```
39. 142 138 154 155 142; 143 156 24 25 157; 144 157 25 31 158; 145 158 31 36 159
40. 146 159 36 41 160; 147 160 41 70 161; 148 161 70 74 162; 149 162 74 79 163
41. 150 163 79 84 164; 151 84 85 165 166; 152 85 86 167 165; 153 86 141 168 167
42. 154 141 142 169 168; 155 142 155 170 169; 156 164 84 166 171
43. 157 172 173 26 24; 158 173 174 16 26; 159 174 175 90 16; 160 175 176 92 90
44. 161 176 177 147 92; 162 178 172 24 156
45. START GROUP DEFINITION
46. ELEMENT
47. _MOMENT 27 31 35 36 39 40 63 64 67 68 71 75 82 86 89 90 93 94 117 118 121 -
48. 122 126 130 135 TO 151 154 TO 157 160 TO 162
49. _SHEAR 39 40 63 64 93 94 117 118 138 139 146 147
50. END GROUP DEFINITION
51. ELEMENT PROPERTY
52. 27 28 31 32 35 36 39 40 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 94 117 -
53. 118 121 122 125 126 129 130 135 TO 162 THICKNESS 0.3
54. SUPPORTS
55. 27 28 31 32 35 36 39 40 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 94 117 -
56. 118 121 122 125 126 129 130 135 TO 161 -
57. 162 PLATE MAT DIRECT Y SUBGRADE 5400 PRINT
```

JOINT	INFLUENCE AREA
13	0.062500
14	0.062500
16	0.062500
24	0.043750
25	0.043750
26	0.062500
27	0.062500
28	0.062500
31	0.043750
32	0.062500
33	0.062500
36	0.043750
37	0.062500
38	0.062500
41	0.043750
42	0.062500
70	0.043750
71	0.062500
74	0.043750
75	0.062500
76	0.062500
79	0.043750
80	0.062500
81	0.062500
84	0.043750
85	0.062500
86	0.062500
90	0.062500
91	0.062500
92	0.043750
93	0.043750
97	0.062500
98	0.043750

101	0.062500
102	0.043750
105	0.062500
106	0.043750
129	0.062500
130	0.043750
133	0.062500
134	0.043750
137	0.062500
138	0.043750
141	0.062500
142	0.043750
147	0.012500
148	0.012500
149	0.012500
150	0.012500
151	0.012500
152	0.012500
153	0.012500
154	0.012500
155	0.012500
156	0.012500
157	0.012500
158	0.012500
159	0.012500
160	0.012500
161	0.012500
162	0.012500
163	0.012500
164	0.012500
165	0.031250
166	0.021875
167	0.031250
168	0.031250
169	0.021875
170	0.006250
171	0.006250
172	0.021875
173	0.031250
174	0.031250
175	0.031250
176	0.021875
177	0.006250
178	0.006250
TOTAL	3.000000

```
58. DEFINE MATERIAL START
59. ISOTROPIC STEEL
60. E 2.09042E+007
61. POISSON 0.3
62. DENSITY 7.83341
63. ALPHA 1.2E-005
64. DAMP 0.03
65. TYPE STEEL
66. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
67. ISOTROPIC CONCRETE
68. E 2.21467E+006
```

```
 69. POISSON 0.17
 70. DENSITY 2.40262
 71. ALPHA 1E-005
 72. DAMP 0.05
 73. TYPE CONCRETE
 74. STRENGTH FCU 2812.28
 75. END DEFINE MATERIAL
 76. MEMBER PROPERTY AMERICAN
 77. 133 134 PRIS YD 0.45 ZD 0.45
 78. CONSTANTS
 79. MATERIAL CONCRETE ALL
 80. LOAD 1 DL
 81. SELFWEIGHT Y -1
 82. ELEMENT LOAD
 83. 27 28 31 32 35 36 39 40 63 64 67 68 71 72 75 76 81 82 85 86 89 90 93 94 117 -
 84. 118 121 122 125 126 129 130 135 TO 162 PR GY -4.1
 85. JOINT LOAD
 86. 145 FY -2
 87. 146 FY -1.5
 88. ******************************************************************
 89. LOAD 2 LL
 90. JOINT LOAD
 91. 145 FY -3.5
 92. 146 FY -2.5
 93. ******************************************************************
 94. LOAD 3 WLX
 95. JOINT LOAD
 96. 146 FX 0.1 FY 0.21
 97. ******************************************************************
 98. LOAD 4 WLZ
 99. JOINT LOAD
100. 145 FY -0.7 FZ 0.5
101. 146 FY 0.35 FZ 0.3
102. ******ALLOWABLE STRESS LOAD COMBS**********************************************
103. LOAD COMB 101 DL + LL
104. 1 1.0 2 1.0
105. LOAD COMB 102 DL + 0.75*LL + 0.75*WLX
106. 1 1.0 2 0.75 3 0.75
107. LOAD COMB 103 DL + 0.75*LL + 0.75*WLZ
108. 1 1.0 2 0.75 4 0.75
109. LOAD COMB 104 0.6*DL + WLX
110. 1 0.6 3 1.0
111. LOAD COMB 105 0.6*DL + WLZ
112. 1 0.6 4 1.0
113. ****ULTIMATE STRENGTH LOAD COMBS*********************************************
114. ******************************************************************************
115. LOAD COMB 200 1.4*DL
116. 1 1.4
117. LOAD COMB 201 1.4D+1.7(LL)
118. 1 1.4 2 1.7
119. LOAD COMB 202 1.2D+1.6WLX+1.0L
120. 1 1.2 2 1.0 3 1.6
121. LOAD COMB 203 1.2D+1.6WLZ+1.0L
122. 1 1.2 2 1.0 4 1.6
123. LOAD COMB 204 0.9D+1.6WLX
124. 1 0.9 3 1.6
```

```
125. LOAD COMB 205 0.9D+1.6WLZ
126. 1 0.9 4 1.6
127. ********************************************************************************
128. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**79**	**NUMBER OF MEMBERS**	**2**
NUMBER OF PLATES	**60**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**77**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 75/ 13/ 42 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 243
TOTAL LOAD COMBINATION CASES = 11 SO FAR.
SIZE OF STIFFNESS MATRIX = 11 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.4/ 163366.6 MB

```
129. LOAD LIST 200 TO 205
130. START CONCRETE DESIGN
```

```
131. CODE ACI 1999
132. TRACK 0 ALL
133. MAXMAIN 25 ALL
134. CLB 0.075 MEMB 133 134
135. CLS 0.075 ALL
136. FC 3058 MEMB 133 134
137. DESIGN COLUMN 133 134
```

==

COLUMN NO. 133 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	200	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

==

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	200	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

138. END CONCRETE DESIGN
139. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 4,2020 TIME= 17:23:18 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3		
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023			Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023		
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD		
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC		
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI		

ATTACHMENT 5 STAAD Input & Output File of Foundation Mkd. F3

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG 29, 2020                *
*           Time=    12:39:27                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: ISOLATED_FDN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 02-APR-19
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 13 0 -4 -1.325; 16 0 -4 -0.25; 24 0.5 -4 -0.25; 25 0.5 -4 -0.5
     9. 26 0.25 -4 -0.25; 27 0.25 -4 -0.5; 28 0 -4 -0.5; 31 0.5 -4 -0.75
    10. 32 0.25 -4 -0.75; 33 0 -4 -0.75; 36 0.5 -4 -1; 37 0.25 -4 -1; 38 0 -4 -1.
    11. 41 0.5 -4 -1.325; 42 0.25 -4 -1.325; 90 -0.25 -4 -0.25; 91 -0.25 -4 -0.5
    12. 92 -0.5 -4 -0.25; 93 -0.5 -4 -0.5; 97 -0.25 -4 -0.75; 98 -0.5 -4 -0.75
    13. 101 -0.25 -4 -1; 102 -0.5 -4 -1; 105 -0.25 -4 -1.325; 106 -0.5 -4 -1.325
    14. 146 0 -1.8 -0.5; 147 -0.825 -4 -0.25; 148 -0.825 -4 -0.5; 149 -0.825 -4 -0.75
    15. 150 -0.825 -4 -1; 151 -0.825 -4 -1.325; 156 0.825 -4 -0.25; 157 0.825 -4 -0.5
    16. 158 0.825 -4 -0.75; 159 0.825 -4 -1; 160 0.825 -4 -1.325; 161 0.5 -4 0
    17. 162 0.25 -4 0; 163 0 -4 0; 164 -0.25 -4 0; 165 -0.5 -4 0; 166 -0.825 -4 0
    18. 167 0.825 -4 0; 168 0.5 -4 0.325; 169 0.25 -4 0.325; 170 0 -4 0.325
    19. 171 -0.25 -4 0.325; 172 -0.5 -4 0.325; 173 -0.825 -4 0.325; 174 0.825 -4 0.325
    20. MEMBER INCIDENCES
    21. 134 28 146
    22. ELEMENT INCIDENCES SHELL
    23. 27 24 26 27 25; 28 26 16 28 27; 31 25 27 32 31; 32 27 28 33 32; 35 31 32 37 36
    24. 36 32 33 38 37; 39 36 37 42 41; 40 37 38 13 42; 81 16 90 91 28; 82 90 92 93 91
    25. 85 28 91 97 33; 86 91 93 98 97; 89 33 97 101 38; 90 97 98 102 101
    26. 93 38 101 105 13; 94 101 102 106 105; 135 92 147 148 93; 136 93 148 149 98
    27. 137 98 149 150 102; 138 102 150 151 106; 143 156 24 25 157; 144 157 25 31 158
    28. 145 158 31 36 159; 146 159 36 41 160; 147 161 162 26 24; 148 162 163 16 26
    29. 149 163 164 90 16; 150 164 165 92 90; 151 165 166 147 92; 152 167 161 24 156
    30. 153 168 169 162 161; 154 169 170 163 162; 155 170 171 164 163
    31. 156 171 172 165 164; 157 172 173 166 165; 158 174 168 161 167
    32. START GROUP DEFINITION
    33. ELEMENT
    34. _MOMENT 27 31 35 36 39 40 82 86 89 90 93 94 135 TO 138 143 TO 158
    35. _SHEAR 39 40 93 94 135 TO 138 143 TO 146 151 TO 158
    36. END GROUP DEFINITION
    37. ELEMENT PROPERTY
    38. 27 28 31 32 35 36 39 40 81 82 85 86 89 90 93 94 135 TO 138 143 TO 157 -
```

```
39. 158 THICKNESS 0.3
40. SUPPORTS
41. 27 28 31 32 35 36 39 40 81 82 85 86 89 90 93 94 135 TO 138 143 TO 157 -
42. 158 PLATE MAT DIRECT Y SUBGRADE 5400 PRINT
```

JOINT	INFLUENCE AREA
13	0.040625
16	0.062500
24	0.071875
25	0.071875
26	0.062500
27	0.062500
28	0.062500
31	0.071875
32	0.062500
33	0.062500
36	0.082656
37	0.071875
38	0.071875
41	0.046719
42	0.040625
90	0.062500
91	0.062500
92	0.071875
93	0.071875
97	0.062500
98	0.071875
101	0.071875
102	0.082656
105	0.040625
106	0.046719
147	0.040625
148	0.040625
149	0.040625
150	0.046719
151	0.026406
156	0.040625
157	0.040625
158	0.040625
159	0.046719
160	0.026406
161	0.082656
162	0.071875
163	0.071875
164	0.071875
165	0.082656
166	0.046719
167	0.046719
168	0.046719
169	0.040625
170	0.040625
171	0.040625
172	0.046719
173	0.026406

```
      174          0.026406
   TOTAL           2.722501
43. DEFINE MATERIAL START
44. ISOTROPIC STEEL
45. E 2.09042E+007
46. POISSON 0.3
47. DENSITY 7.83341
48. ALPHA 1.2E-005
49. DAMP 0.03
50. TYPE STEEL
51. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
52. ISOTROPIC CONCRETE
53. E 2.21467E+006
54. POISSON 0.17
55. DENSITY 2.40262
56. ALPHA 1E-005
57. DAMP 0.05
58. TYPE CONCRETE
59. STRENGTH FCU 2812.28
60. END DEFINE MATERIAL
61. MEMBER PROPERTY AMERICAN
62. 134 PRIS YD 0.45 ZD 0.45
63. CONSTANTS
64. MATERIAL CONCRETE ALL
65. LOAD 1 DL
66. SELFWEIGHT Y -1 LIST ALL
67. ELEMENT LOAD
68. 27 28 31 32 35 36 39 40 81 82 85 86 89 90 93 94 135 TO 138 143 TO 157 -
69. 158 PR GY -4.1
70. JOINT LOAD
71. 146 FX 1 FY -2
72. ******************************************************************************
73. LOAD 2 LL
74. JOINT LOAD
75. 146 FX 2.2 FY -3.7 FZ -0.2
76. ******************************************************************************
77. LOAD 3 WLX
78. JOINT LOAD
79. 146 FX 0.2 FY 0.2
80. ******************************************************************************
81. LOAD 4 WLZ
82. JOINT LOAD
83. 146 FX -0.1 FY -0.1 FZ -0.2
84. ******ALLOWABLE STRESS LOAD COMBS************************************************
85. LOAD COMB 101 DL + LL
86. 1 1.0 2 1.0
87. LOAD COMB 102 DL + 0.75*LL + 0.75*WLX
88. 1 1.0 2 0.75 3 0.75
89. LOAD COMB 103 DL + 0.75*LL + 0.75*WLZ
90. 1 1.0 2 0.75 4 0.75
91. LOAD COMB 104 0.6*DL + WLX
92. 1 0.6 3 1.0
93. LOAD COMB 105 0.6*DL + WLZ
94. 1 0.6 4 1.0
95. ****ULTIMATE STRENGTH LOAD COMBS*********************************************
96. ******************************************************************************
```

```
 97. LOAD COMB 200 1.4*DL
 98. 1 1.4
 99. LOAD COMB 201 1.4D+1.7(LL)
100. 1 1.4 2 1.7
101. LOAD COMB 202 1.2D+1.6WLX+1.0L
102. 1 1.2 2 1.0 3 1.6
103. LOAD COMB 203 1.2D+1.6WLZ+1.0L
104. 1 1.2 2 1.0 4 1.6
105. LOAD COMB 204 0.9D+1.6WLX
106. 1 0.9 3 1.6
107. LOAD COMB 205 0.9D+1.6WLZ
108. 1 0.9 4 1.6
109. ********************************************************************************
110. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**50**	**NUMBER OF MEMBERS**	**1**
NUMBER OF PLATES	**36**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**49**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 40/ 13/ 45 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 153
TOTAL LOAD COMBINATION CASES = 11 SO FAR.
SIZE OF STIFFNESS MATRIX = 7 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 163627.0 MB

```
111. LOAD LIST 200 TO 205
112. START CONCRETE DESIGN
```

```
113. CODE ACI 1999
114. TRACK 0 ALL
115. MAXMAIN 25 ALL
116. CLB 0.075 MEMB 134
117. CLS 0.075 ALL
118. FC 3058 MEMB 134
119. DESIGN COLUMN 134
```

==========

COLUMN NO. 134 DESIGN PER ACI 318-99 - AXIAL + BENDING

FY - 413.7 FC - 30.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
20 - 12 MM	1.117	200	END	0.700

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 12 SPACING 192.00 MM

**********END OF COLUMN DESIGN RESULTS**********

121. END CONCRETE DESIGN
122. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 29,2020 TIME= 12:39:37 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2014 Bentley Systems, Inc.
*                http://www.bentley.com                     *
*************************************************************
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01			Outotec / L&T Revision No.:	0	1	2	3
Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023			Date	29-08-2020	03-10-2020	19-11-2020	17-11-2023
Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT			Revision Text	FIRST ISSUE	REVISED AS PER EMPLOYER COMMENTS ON 29-09-20 & REVISED AS MARKED	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD	SHD			Prepared By	SHD	SHD	SHD	SHD
Checked	SGC	SGC	SGC	SGC			Checked	SGC	SGC	SGC	SGC
Approved	PRI	PRI	PRI	PRI			Approved	PRI	PRI	PRI	PRI

ATTACHMENT 6 STAAD Input & Output File of WEIGHFEEDER SUPPORT STRUCTURE

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries5          *
*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    AUG 29, 2020                *
*           Time=    14:41:49                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: 3360_PLATFORM_2225 _updated_latest.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 30-MAY-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0.5 0; 2 5.05 0.5 0; 3 0 0.5 5; 4 5.05 0.5 5; 5 0 2.9 0; 6 5.05 2.9 0
     9. 7 0 2.9 5; 8 5.05 2.9 5; 9 1.7 0.5 0; 10 1.7 2.9 0; 11 3.2 0.5 0; 12 3.2 2.9 0
    10. 13 1.7 2.9 5; 14 3.2 2.9 5; 15 0 2.9 3.93; 16 1.7 2.9 3.93; 17 3.2 2.9 3.93
    11. 18 5.05 2.9 3.93; 19 2.45 2.9 5; 20 0 2.9 1.965; 21 5.05 2.9 1.965
    12. 22 1.7 2.9 1.965; 23 3.2 2.9 1.965; 24 0 2.9 -1.12; 25 5.05 2.9 -1.12
    13. 26 1.7 2.9 -1.12; 27 3.2 2.9 -1.12; 28 0 1.78 0; 29 5.05 1.78 0; 30 1.7 1.78 0
    14. 31 3.2 1.78 0; 32 1.7 2.9 2.73; 33 3.2 2.9 2.73; 34 1.7 2.9 1.63
    15. 35 3.2 2.9 1.63; 36 1.7 2.9 0.53; 37 3.2 2.9 0.53; 38 1.7 2.9 -0.57
    16. 39 3.2 2.9 -0.57; 40 0 2.9 0.95; 41 1.7 2.9 0.95; 42 3.2 2.9 0.95
    17. 43 5.05 2.9 0.95; 44 0 2.9 2.915; 45 1.7 2.9 2.915; 46 5.05 2.9 2.915
    18. 47 3.2 2.9 2.915
    19. MEMBER INCIDENCES
    20. 1 1 28; 2 3 7; 3 4 8; 4 2 29; 5 5 40; 6 6 43; 7 7 13; 8 9 30; 9 11 31
    21. 10 13 19; 11 10 36; 12 14 8; 13 12 37; 14 5 10; 15 12 6; 16 15 7; 17 16 13
    22. 18 15 16; 19 17 14; 20 16 17; 21 18 8; 22 17 18; 23 7 16; 24 19 14; 25 16 19
    23. 26 19 17; 27 17 8; 28 20 44; 29 10 20; 30 20 16; 31 21 46; 32 12 21; 33 21 17
    24. 34 22 32; 35 20 22; 36 23 33; 37 21 23; 38 28 5; 39 29 6; 40 30 10; 41 31 12
    25. 42 5 24; 43 24 28; 44 10 38; 45 26 30; 46 6 25; 47 25 29; 48 12 39; 49 27 31
    26. 50 24 26; 51 27 25; 52 10 24; 53 12 25; 54 19 3; 55 19 4; 56 28 30; 57 31 29
    27. 58 10 28; 59 28 9; 60 12 29; 61 29 11; 62 32 45; 63 33 47; 64 34 22; 65 35 23
    28. 66 36 41; 67 37 42; 68 38 26; 69 39 27; 70 40 20; 71 41 34; 72 40 41; 73 42 35
    29. 74 43 21; 75 42 43; 76 44 15; 77 45 16; 78 44 45; 79 46 18; 80 47 17; 81 46 47
    30. START GROUP DEFINITION
    31. FLOOR
    32. _FL1 5 TO 7 10 TO 37 42 44 46 48 50 TO 53 62 TO 81
    33. END GROUP DEFINITION
    34. DEFINE MATERIAL START
    35. ISOTROPIC STEEL
    36. E 2.09042E+007
    37. POISSON 0.3
    38. DENSITY 7.83341
```

```
39. ALPHA 1.2E-005
40. DAMP 0.03
41. TYPE STEEL
42. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
43. END DEFINE MATERIAL
44. MEMBER PROPERTY EUROPEAN
45. 1 TO 4 38 39 TABLE TB IPE200 WP 0.12 TH 0.012 BW 0.12 BT 0.012
46. 7 10 TO 13 17 19 24 34 36 62 TO 67 71 73 77 80 TABLE ST IPE300
47. 42 44 46 48 68 69 TABLE ST UPN200
48. 20 50 51 TABLE ST UPN200
49. 14 15 18 22 35 37 72 75 78 81 TABLE ST UPN100
50. 23 25 TO 27 29 30 32 33 43 45 47 49 52 53 TABLE ST L65X65X6
51. 56 TO 61 TABLE ST L75X75X8
52. 54 55 TABLE ST L90X90X8
53. 5 6 16 21 28 31 70 74 76 79 TABLE ST UPN240
54. 8 9 40 41 TABLE ST IPE300
55. CONSTANTS
56. BETA 90 MEMB 1 TO 4 8 9 38 TO 41
57. MATERIAL STEEL ALL
58. MEMBER RELEASE
59. 7 11 13 TO 15 18 20 22 35 37 56 57 72 75 78 81 START MY MZ
60. 12 14 15 17 TO 20 22 35 37 56 57 72 75 78 81 END MY MZ
61. MEMBER TRUSS
62. 23 25 TO 27 29 30 32 33 43 45 47 49 52 TO 55 58 TO 61
63. SUPPORTS
64. 1 TO 4 9 11 PINNED
65. DEFINE WIND LOAD
```

***** NOTE: If any floor diaphragm is present in the model Wind Load definition should be defined after Floor Diaphragm definition. Otherwise wind load generation may be unsuccessful during analysis.**

```
66. TYPE 1 WIND 1
67. INT 0.07 HEIG  5
68. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD
69. SELFWEIGHT Y -1
70. MEMBER LOAD
71. 5 TO 7 10 TO 22 24 28 31 34 TO 37 42 44 46 48 50 51 62 TO 81 UNI GY -0.06
72. *LOAD FROM 3360-FDR-001
73. JOINT LOAD
74. 10 FY 0.5
75. 12 FY 0.5
76. 34 FY -2.3
77. 35 FY -1.4
78. ************************************************************
79. LOAD 2 LOADTYPE NONE  TITLE LL
80. MEMBER LOAD
81. 5 6 10 11 13 16 17 19 TO 21 24 28 31 34 36 42 44 46 48 62 TO 71 73 74 76 77 -
82. 79 80 UNI GY -0.425
83. *LOAD FROM 3360-FDR-001
84. JOINT LOAD
85. 10 FY 0.5
86. 12 FY 0.5
87. 34 FY -3.5
88. 35 FY -2.1
89. **********************************
```

```
 90. LOAD 3 LOADTYPE NONE  TITLE WLX
 91. WIND LOAD X 2 TYPE 1 XR 0 5 YR 0 3 ZR -1.5 5 OPEN
 92. LOAD 4 LOADTYPE NONE  TITLE WLZ
 93. WIND LOAD Z 2 TYPE 1 XR 0 5 YR 0 5 ZR -1.5 5 OPEN
 94. LOAD COMB 10 1.4*DL
 95. 1 1.4
 96. LOAD COMB 11 1.2*DL + 1.6*LL
 97. 1 1.2 2 1.6
 98. LOAD COMB 12 1.2*DL + 1.6*LL + 0.8*WLX
 99. 1 1.2 2 1.6 3 0.8
100. LOAD COMB 13 1.2*DL + 1.0*LL + 1.6*WLX
101. 1 1.2 2 1.0 3 1.6
102. LOAD COMB 14 1.2*DL + 1.6*LL + 0.8*WLZ
103. 1 1.2 2 1.6 4 0.8
104. LOAD COMB 15 1.2*DL + 1.0*LL + 1.6*WLZ
105. 1 1.2 2 1.0 4 1.6
106. LOAD COMB 16 0.9*DL + 1.6*WLX
107. 1 0.9 3 1.6
108. LOAD COMB 17 0.9*DL + 1.6*WLZ
109. 1 0.9 4 1.6
110. ****LOAD COMBS FOR DEFLECTION**************************************************
111. LOAD COMB 21 DL + LL
112. 1 1.0 2 1.0
113. LOAD COMB 22 DL + LL + WLX
114. 1 1.0 2 1.0 3 1.0
115. LOAD COMB 23 DL + LL + WLZ
116. 1 1.0 2 1.0 4 1.0
117. LOAD COMB 24 DL + WLX
118. 1 1.0 3 1.0
119. LOAD COMB 25 DL + WLZ
120. 1 1.0 4 1.0
121. ************************************************88
122. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**47**	**NUMBER OF MEMBERS**	**81**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 37/ 9/ 60 DOF

TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 264
TOTAL LOAD COMBINATION CASES = 13 SO FAR.
SIZE OF STIFFNESS MATRIX = 16 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.3/ 163626.2 MB

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE NONE TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.240988651E+01
Y = 0.277850923E+01
Z = 0.214327006E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -8.22
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 17.61 MY= 0.00 MZ= -19.80

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8.22
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -17.61 MY= 0.00 MZ= 19.80

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	9.72079E-04	27
Y =	-3.93683E-01	22
Z =	-5.38171E-02	27
RX=	2.43819E-03	36
RY=	7.28200E-06	24
RZ=	1.39556E-04	25

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE NONE TITLE LL

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.240946618E+01
Y = 0.290000008E+01
Z = 0.215029395E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -16.28
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 35.00 MY= 0.00 MZ= -39.22

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 16.28
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -35.00 MY= 0.00 MZ= 39.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	1.90279E-03	27
Y =	-7.12787E-01	32
Z =	-9.43491E-02	27
RX=	4.33505E-03	36
RY=	1.64399E-05	24
RZ=	-2.27297E-04	24

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE WLX

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.157830664E+01
Y = 0.250270183E+01
Z = 0.177212965E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 1.24
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 2.20 MZ= -3.11

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = -1.24
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -2.20 MZ= 3.11

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	1.72707E-02	40
Y =	2.07047E-03	27
Z =	4.86805E-03	8
RX=	2.38842E-05	4
RY=	-6.75822E-05	30
RZ=	-1.78582E-04	3

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE WLZ

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.130529316E+01
Y = 0.245786455E+01
Z = 0.218961855E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.77

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.90 MY= -1.01 MZ= 0.00

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -0.77

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -1.90 MY= 1.01 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-7.80618E-03	25
Y =	8.18592E-02	26
Z =	1.77316E-01	26
RX=	8.78806E-04	1
RY=	6.14862E-05	19
RZ=	9.61747E-05	24

********** END OF DATA FROM INTERNAL STORAGE **********

```
123. LOAD LIST 1 TO 4
124. PRINT SUPPORT REACTION LIST 3 4 9 11
```

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
3	1	1.07	1.83	-0.05	0.00	0.00	0.00
	2	2.20	3.65	-0.20	0.00	0.00	0.00
	3	-0.24	-0.20	0.00	0.00	0.00	0.00
	4	-0.07	0.09	-0.20	0.00	0.00	0.00
4	1	-1.07	1.69	-0.06	0.00	0.00	0.00
	2	-2.19	3.36	-0.20	0.00	0.00	0.00
	3	-0.21	0.20	0.00	0.00	0.00	0.00
	4	-0.07	0.21	-0.15	0.00	0.00	0.00
9	1	0.02	1.79	-0.06	0.00	0.00	0.00
	2	0.04	3.40	-0.11	0.00	0.00	0.00
	3	-0.50	0.59	0.00	0.00	0.00	0.00
	4	0.00	0.00	-0.04	0.00	0.00	0.00
11	1	-0.02	1.21	-0.06	0.00	0.00	0.00
	2	-0.04	2.46	-0.11	0.00	0.00	0.00
	3	-0.27	-0.29	0.00	0.00	0.00	0.00
	4	0.14	0.18	-0.03	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

```
125. LOAD LIST 10 TO 17
126. PARAMETER 1
127. CODE LRFD
128. KZ 0.85 MEMB 23 25 TO 27 29 30 32 33 43 45 47 49 52 53 58 TO 61
129. KY 1.08 MEMB 54 55
130. KZ 0.3 MEMB 54 55
131. KY 0.85 MEMB 23 25 TO 27 29 30 32 33 52 53 58 TO 61
132. TRACK 0 ALL
133. RATIO 1 ALL
134. CHECK CODE ALL
```

STAAD.Pro CODE CHECKING - (LRFD 3RD EDITION) v1.0
****

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.088	11
		3.66 C	0.01	-0.81	1.28
2	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.132	14
		3.30 C	0.00	1.23	2.40
3	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.129	14
		2.94 C	0.00	1.22	2.40
4	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.088	12
		3.91 C	-0.01	-0.81	1.28
5	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.269	12
		0.61 C	-0.01	2.14	0.00
6	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.267	12
		0.64 C	0.00	2.13	0.00
7	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.143	12
		0.17 C	0.00	-2.01	1.70
8	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.064	12
		7.76 C	0.02	0.32	1.28
9	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.052	14
		5.45 C	-0.01	0.34	1.28
10	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.167	14
		0.09 T	0.00	1.42	0.75
11	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.290	12
		0.23 T	0.01	-4.08	0.53
12	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.108	14
		0.02 C	0.01	-1.49	0.00
13	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.206	12
		0.07 T	0.01	-2.91	0.53
14	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.059	15
		0.02 T	-0.01	-0.03	0.85
15	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.042	10
		0.00 T	0.00	-0.04	0.92

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
16 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.155	15
		0.28 C	0.00	1.25	1.07
17 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.372	12
		0.01 T	-0.01	-5.27	0.00
18 ST	UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.058	15
		0.00 C	-0.01	-0.03	0.85
19 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.289	12
		0.00 C	0.00	-4.12	0.00
20 ST	UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.052	14
		0.09 C	-0.01	-0.22	0.75
21 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.152	14
		0.53 C	0.00	1.22	1.07
22 ST	UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.042	10
		0.00 C	0.00	-0.04	0.92
23 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.045	17
		0.30 C	0.00	0.00	0.00
24 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.144	14
		0.03 C	0.00	1.42	0.00
25 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.030	15
		0.34 C	0.00	0.00	1.31
26 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.026	17
		0.44 T	0.00	0.00	1.31
27 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.048	16
		0.29 C	0.00	0.00	0.00
28 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.155	12
		0.46 C	0.00	-1.22	0.87
29 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.079	16
		0.31 C	0.00	0.00	0.00
30 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.066	13
		0.26 C	0.00	0.00	2.60

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
31	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.154	12
		0.35 C	0.00	-1.23	0.87
32	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.079	17
		0.29 C	0.00	0.00	0.00
33	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.041	12
		0.15 C	0.00	0.00	0.00
34	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.784	12
		0.23 T	-0.01	-11.14	0.00
35	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.058	15
		0.00 C	-0.01	-0.03	0.85
36	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.554	12
		0.07 T	-0.01	-7.87	0.00
37	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.042	10
		0.00 T	0.00	-0.04	0.92
38	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.132	12
		3.55 C	0.00	-1.40	1.12
39	TB IPE200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.133	11
		3.57 C	0.00	-1.40	1.12
40	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.062	12
		7.30 C	0.02	0.32	0.00
41	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.049	14
		4.99 C	-0.01	0.34	0.00
42	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.149	12
		0.09 C	0.01	0.74	0.00
43	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.027	17
		0.26 C	0.00	0.00	1.58
44	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.023	11
		0.44 T	0.00	0.10	0.00
45	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.067	14
		0.64 C	0.00	0.00	1.58

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
46 ST	UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.147	12
		0.12 C	0.00	0.74	0.00
47 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.024	17
		0.23 C	0.00	0.00	1.58
48 ST	UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.024	12
		0.44 T	0.00	0.11	0.00
49 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.066	14
		0.63 C	0.00	0.00	1.58
50 ST	UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.017	15
		0.01 C	-0.01	-0.04	0.85
51 ST	UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.011	13
		0.04 C	0.00	-0.04	0.92
52 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.016	13
		0.11 C	0.00	0.00	0.00
53 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.002	16
		0.04 T	0.00	0.00	0.00
54 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.378	11
		6.73 C	0.00	0.00	3.43
55 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.394	12
		6.76 C	0.00	0.00	3.54
56 ST	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.025	15
		0.02 T	-0.01	0.00	0.85
57 ST	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.020	13
		0.10 C	0.00	0.00	0.92
58 ST	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.024	16
		0.64 T	0.00	0.00	2.04
59 ST	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.080	16
		0.94 C	0.00	0.00	2.13
60 ST	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.040	13
		0.45 C	0.00	0.00	2.16

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
61	ST L75X75X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.023	17
		0.25 C	0.00	0.00	2.25
62	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.650	12
		0.23 T	0.01	-9.25	0.00
63	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.478	12
		0.07 T	0.00	-6.81	0.00
64	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.825	12
		0.23 T	0.00	-11.79	0.00
65	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.571	14
		0.23 T	0.00	-8.16	0.00
66	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.506	12
		0.23 T	0.01	-7.16	0.42
67	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.355	12
		0.07 T	0.00	-5.05	0.42
68	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.021	12
		0.45 T	0.00	-0.08	0.09
69	ST UPN200		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.020	12
		0.44 T	0.00	-0.07	0.09
70	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.128	14
		0.58 C	0.00	-1.02	1.02
71	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.825	12
		0.23 T	0.00	-11.79	0.68
72	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.058	15
		0.00 C	-0.01	-0.03	0.85
73	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.571	14
		0.23 T	0.00	-8.16	0.68
74	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.128	14
		0.51 C	0.00	-1.02	1.02
75	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.042	10
		0.00 C	0.00	-0.04	0.92

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
76	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.154	12
		0.46 C	0.00	-1.22	0.00
77	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.614	12
		0.23 T	0.01	-8.72	0.00
78	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.058	15
		0.00 C	-0.01	-0.03	0.85
79	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.154	12
		0.35 C	0.00	-1.22	0.00
80	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.455	12
		0.07 T	0.00	-6.48	0.00
81	ST UPN100		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.042	10
		0.00 C	0.00	-0.04	0.92

********** END OF TABULATED RESULT OF DESIGN **********

135. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 29,2020 TIME= 14:42: 0 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
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*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
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